Online Supplement to
Financial Strategy Library Compression under Partial Information: Resource Constraints and Generative Value

David Ramirez
ORCID: 0009-0000-3128-5123
Corresponding email: ramirezdavv@gmail.com

Release v0.1.0-preprint — August 23, 2026

Contents

This supplement collects the extended theoretical material and complete numerical records omitted from the primary appendix. The notation, finite-state model, timing, assumptions, and evidence classifications are those of the main paper. References to the main paper are descriptive so that this document compiles independently; all numbered cross-references below are internal to the supplement.

S1 Belief Occupation and Coverage

The main paper's operational–generative value decomposition separates the passive operating channel from the research premium. This section develops the passive channel through discounted belief occupation and then gives the associated finite coverage, threshold, and belief-dynamics results. These are supporting results: a one-shot cost-covering region is not an optimized Bellman research region.

Throughout, expectations and occupation rows are conditional on the displayed initial belief. Every horizon and state carrier is finite. Finiteness justifies the sum rearrangements and attained extrema below; no limiting argument, closure-detectability condition, stationary contraction, or path–outcome independence is used.

S1.1 Discounted gap sums and operational insertion

Definition S1.1 (Positive gap and discounted gap sum)**.** For a fixed candidate s, define

$$\begin{aligned} \Delta_{s,L}(b) &:= \max\{j_s(b) - F_L(b), 0\}, \\ G_0(b; L, s) &= 0, \\ G_{n+1}(b; L, s) &= \Delta_{s,L}(b) + \beta \sum_{b'} P_B(b, b') G_n(b'; L, s). \end{aligned} \tag{S1.1}$$

Let

$$w_n(b, b') := \sum_{t=0}^{n-1} \beta^t \Pr(B_t = b' \mid B_0 = b)$$

be discounted expected occupation, not a normalized probability.

For the supporting frozen-library process, set $\Delta_n^{\mathrm{op,aux}}(s \mid b, L) := P_n(b, L \cup \{s\}) - P_n(b, L)$, where P_n is its passive value.

Proposition S1.2 (Passive gap-sum identity)**.** *For the supporting frozen-library process,*

$$\Delta_n^{\mathrm{op,aux}}(s \mid b, L) = G_n(b; L, s) = \sum_{b'} w_n(b, b') \Delta_{s,L}(b'). \tag{S1.2}$$

When the unified process has this frozen-library no-research counterfactual, combining the occupancy identity with the operational–generative value decomposition in the main paper gives

$$\mathcal{I}_n = \sum_{b'} w_n(b, b') \Delta_{s,L}(b') + \Delta_n^{\mathrm{gen}}. \tag{S1.3}$$

The occupancy formula shows why a zero current gap does not make a candidate worthless if the belief process later visits states with positive gaps.

Corollary S1.3 (Zero current gap, positive future operating value)**.** *With a deterministic move to a gap-two belief, $n = 2$, and $\beta = 1/2$, operational insertion value is one.*

Proposition S1.4 (Diminishing marginal operational innovation). *If $L \subseteq L'$, then every fixed candidate satisfies*

$$\Delta_n^{\mathrm{op}}(s \mid b, L') \le \Delta_n^{\mathrm{op}}(s \mid b, L).$$

For Proposition S1.2, insertion replaces the frontier pointwise by $\max\{F_L, j_s\}$. Subtracting F_L gives $\Delta_{s,L}$. Conditional on the displayed initial belief and frozen library, subtracting the two passive recursions and inducting on the horizon gives (S1.1). Unrolling that finite recursion gives the discounted-occupancy representation; only finite sums are rearranged. The delayed-coverage specialization in Corollary S1.3 follows by placing unit discounted mass on the future belief with gap two. For Proposition S1.4, library inclusion raises the frontier and lowers the pointwise positive gap, and nonnegative discounted expectation preserves the order.

Persistence itself is ambiguous because it may keep the process in, or away from, the candidate's advantage region. The valid positive comparison is occupation alignment on positive-gap beliefs, developed below, not persistence in isolation.

S1.2 Coverage potential and occupation bounds

Coverage links a candidate's operational advantage to the beliefs at which that advantage will matter.

Definition S1.5 (Certified gap and coverage potential). For a fixed project q, library K, candidate profile j_q, and operational frontier F_K, define the certified gap

$$\Delta(q, K, b) := \max\{j_q(b) - F_K(b), 0\}.$$

Let $\omega_t(b, b') \ge 0$ be the date-t occupation weight of future belief b' from initial belief b. For horizon H, discount β, and candidate-survival factor ρ_q, set

$$W_H^{\beta,\rho_q}(b, b') := \sum_{t<H} \beta^t \rho_q^t \omega_t(b, b'), \tag{S1.4}$$

$$\Psi_H(q, K, b) := \sum_{b' \in B} W_H^{\beta,\rho_q}(b, b') \Delta(q, K, b'). \tag{S1.5}$$

The potential Ψ_H is discounted exposure to beliefs at which the candidate improves the current frontier. The weight W_H^{β,ρ_q} need not itself be a probability: it combines belief occupation, discounting, and survival of the candidate's advantage.

Theorem S1.6 (Exact finite coverage representation). *For every finite horizon, occupation table, gap table, project, library, and initial belief,*

$$\Psi_H(q, K, b) = \sum_{t<H} \sum_{b' \in B} \beta^t \rho_q^t \omega_t(b, b') \Delta(q, K, b'). \tag{S1.6}$$

The right-hand side is the date-by-date one-shot gross operational value of the supplied candidate.

With nonnegative inputs, the same finite order argument makes Ψ_H increasing in the gap, discount, survival, and occupation weights. A pointwise higher frontier lowers the positive-part gap of a fixed candidate and therefore weakly lowers its coverage potential.

Corollary S1.7 (Finite occupation bounds). *For any nonempty $A \subseteq B$, its attained minimum gap times its discounted occupation is a lower bound for Ψ_H. The attained global maximum gap times total discounted occupation is an upper bound. If the gap vanishes outside A, the upper bound may use occupation restricted to A.*

Example S1.8 (Delayed coverage)**.** On a two-belief grid, let date zero occupy the current belief and date one the other belief. With $H = 2$, $\beta = 1/2$, $\rho_q = 1$, current gap zero, and future gap two,

$$\Delta(q, K, b_{\text{current}}) = 0, \qquad \Psi_2(q, K, b_{\text{current}}) = 1 > 0.$$

Screening only at the current belief would discard this future advantage.

Fix (q, K, b, H). Substitute (S1.4) into (S1.5). Since both index sets are finite, distributivity and finite-sum commutation give

$$\begin{aligned}\Psi_H(q, K, b) &= \sum_{b' \in B} \left[\sum_{t<H} \beta^t \rho_q^t \omega_t(b, b') \right] \Delta(q, K, b') \\ &= \sum_{b' \in B} \sum_{t<H} \beta^t \rho_q^t \omega_t(b, b') \Delta(q, K, b') \\ &= \sum_{t<H} \sum_{b' \in B} \beta^t \rho_q^t \omega_t(b, b') \Delta(q, K, b'),\end{aligned}$$

which is (S1.6). This proof uses no limiting argument.

For a nonempty region $A \subseteq B$, write

$$\underline{\Delta}_A := \min_{b' \in A} \Delta(q, K, b'), \qquad \overline{\Delta} := \max_{b' \in B} \Delta(q, K, b').$$

All weights and gaps are nonnegative. Hence

$$\underline{\Delta}_A \sum_{b' \in A} W_H^{\beta, \rho_q}(b, b') \leq \Psi_H(q, K, b) \leq \overline{\Delta} \sum_{b' \in B} W_H^{\beta, \rho_q}(b, b').$$

If the gap vanishes outside A, the last sum can be restricted to A. The four supporting monotonicities follow term by term: raising a nonnegative gap, discount, survival factor, or occupation weight raises every affected summand. If $F_0 \leq F_1$, then $\max\{j_q - F_1, 0\} \leq \max\{j_q - F_0, 0\}$, proving frontier saturation for a fixed candidate.

S1.3 Monotone one-shot coverage

For a one-step project, let $p(b) \geq 0$ scale survival or admission and let $\kappa(b)$ be initiation cost.

Definition S1.9 (One-shot cost-covering region)**.** On a finite linearly ordered belief grid with exact transition kernel P, define

$$G(b) := \beta p(b) \sum_{b' \in B} P(b, b') \Delta(b'), \qquad \mathcal{C} := \{b \in B : \kappa(b) \leq G(b)\}.$$

The set $\mathcal{C}$ is the *one-shot cost-covering region.*

Theorem S1.10 (Monotone-gap upper-threshold theorem)**.** *Assume that B is a nonempty finite linearly ordered grid; P is an exact nonnegative row-stochastic kernel that is first-order stochastically monotone; Δ is nonnegative and increasing; p is nonnegative and increasing; $\beta \geq 0$; and κ is antitone. Then G is increasing and $\mathcal{C}$ is an upper set. Consequently, either $\mathcal{C} = \varnothing$, or there is $b^* \in B$ such that*

$$\mathcal{C} = \{b \in B : b^* \leq b\}. \tag{S1.7}$$

In the latter case the region is order-connected on the finite grid.

The theorem is about the displayed one-shot cost-covering region. We reserve *optimal research region* for the action selected by the Bellman policy. Continuation-value differences need not inherit the order properties used above, so the theorem does not characterize that Bellman region.

For each current belief b, let

$$E_\Delta(b) := \sum_{b' \in B} P(b, b')\Delta(b').$$

Here $E_\Delta(b)$ is the conditional expected gap after one transition. First-order stochastic monotonicity of P and monotonicity of Δ make it increasing. Exact row stochasticity and $\Delta \geq 0$ make it nonnegative. If $p \geq 0$ is increasing, then for $b \leq \widetilde{b}$,

$$p(b)E_\Delta(b) \leq p(\widetilde{b})E_\Delta(b) \leq p(\widetilde{b})E_\Delta(\widetilde{b}).$$

Multiplication by $\beta \geq 0$ therefore makes G increasing.

Suppose $b \in \mathcal{C}$ and $b \leq \widetilde{b}$. Antitonicity of cost and monotonicity of gross value yield

$$\kappa(\widetilde{b}) \leq \kappa(b) \leq G(b) \leq G(\widetilde{b}),$$

so $\widetilde{b} \in \mathcal{C}$. Thus $\mathcal{C}$ is an upper set. A nonempty upper set in a finite linear order has a least element b^* and equals $\{b : b^* \leq b\}$; the empty case is separate.

The cutoff directions and the exact failures of the threshold assumptions are collected in Section S3.

S1.4 Patience, survival, and belief dynamics

Theorem S1.11 (Finite patience–survival complementarity)**.** *Let P be an exact finite row-stochastic matrix, $g \geq 0$, and*

$$\Psi_H(\beta, \rho) := \sum_{t=0}^{H-1} (\beta\rho)^t P^t g.$$

The potential is pointwise increasing in either nonnegative β or ρ. If $0 \leq \beta_0 \leq \beta_1$ and $0 \leq \rho_0 \leq \rho_1$, then, componentwise,

$$\Psi_H(\beta_1, \rho_1) + \Psi_H(\beta_0, \rho_0) \geq \Psi_H(\beta_1, \rho_0) + \Psi_H(\beta_0, \rho_1). \tag{S1.8}$$

Conditional on an initial belief, row stochasticity implies that P^t is nonnegative, so $P^t g \geq 0$. Monotonicity in either nonnegative scalar follows term by term. For the four displayed corners,

$$\begin{aligned} &\Psi_H(\beta_1, \rho_1) + \Psi_H(\beta_0, \rho_0) - \Psi_H(\beta_1, \rho_0) - \Psi_H(\beta_0, \rho_1) \\ &\quad = \sum_{t<H} (\beta_1^t - \beta_0^t)(\rho_1^t - \rho_0^t)P^t g \geq 0. \end{aligned} \tag{S1.9}$$

Indeed, the date-t coefficient factorizes exactly as

$$(\beta_1\rho_1)^t + (\beta_0\rho_0)^t - (\beta_1\rho_0)^t - (\beta_0\rho_1)^t = (\beta_1^t - \beta_0^t)(\rho_1^t - \rho_0^t).$$

Every factor is nonnegative. Finiteness of H justifies the displayed termwise expansion. No independence, inverse, infinite series, real derivative, detectability, or contraction enters the result.

A scalar persistence parameter does not order coverage. Consider

$$P(\theta) = \begin{pmatrix} \theta & 1-\theta \\ 1-\theta & \theta \end{pmatrix}, \qquad 0 \leq \theta \leq 1.$$

Proposition S1.12 (Opposite persistence effects)**.** *Let $H = 2$, effective discount $\alpha = 1/2$, initial belief* 0*, and compare $\theta_0 = 1/4$ with $\theta_1 = 3/4$. Exact evaluation gives*

g	$\Psi_2(\theta_0)$	$\Psi_2(\theta_1)$
$(1,0)$	$9/8$	$11/8$
$(0,1)$	$3/8$	$1/8$
$(1,1)$	$3/2$	$3/2.$

Thus greater persistence can raise, lower, or leave coverage unchanged.

Persistence raises coverage when it retains occupation at an uncovered current belief, but lowers it when it traps the process away from uncovered beliefs elsewhere. The positive order must therefore be stated in terms of where occupation moves. Define

$$W^P_{H,\alpha}(b,x) := \sum_{t=0}^{H-1} \alpha^t (P^t)(b,x).$$

Theorem S1.13 (Advantage-region occupation alignment)**.** *Let P_0, P_1 be finite row-stochastic kernels and $g \geq 0$. If, for every initial belief b and every x with $g(x) > 0$,*

$$W^{P_1}_{H,\alpha}(b,x) \geq W^{P_0}_{H,\alpha}(b,x),$$

then

$$\sum_{t=0}^{H-1} \alpha^t P_1^t g \geq \sum_{t=0}^{H-1} \alpha^t P_0^t g$$

componentwise.

Fix the initial belief b. Finite sum exchange yields

$$\begin{aligned}
\Psi^P_{H,\alpha}(b) &= \sum_{t<H} \alpha^t \sum_x (P^t)(b,x) g(x) \\
&= \sum_x \left[\sum_{t<H} \alpha^t (P^t)(b,x) \right] g(x) \\
&= \sum_x W^P_{H,\alpha}(b,x) g(x).
\end{aligned}$$

Outside the advantage region $g(x) = 0$. Inside it, the assumed occupation difference and the gap are nonnegative. Termwise comparison proves Theorem S1.13.

For the exact persistence boundary, $H = 2$ gives $\Psi_2 = g + \alpha P(\theta) g$. At initial belief zero and $\alpha = 1/2$,

$$\begin{aligned}
g = (1,0) &\quad:\quad \Psi_2 = 1 + \theta/2, \\
g = (0,1) &\quad:\quad \Psi_2 = (1-\theta)/2, \\
g = (1,1) &\quad:\quad \Psi_2 = 3/2.
\end{aligned}$$

Substitution of $\theta = 1/4$ and $3/4$ gives respectively $9/8 \to 11/8$, $3/8 \to 1/8$, and $3/2 \to 3/2$. Both matrices are exactly row stochastic. The sign changes because persistence reallocates occupation; it does not order occupation on the positive-gap region without an alignment condition.

S2 Frontier–Closure Interaction

The main paper's compressed state has two economically different margins. A higher frontier improves current operation and makes further operational improvement harder. A richer closure expands the menu of policies that research can generate. This section gives the complete finite interaction theory, its primitive sufficient conditions, and the exact mechanisms that make the sign ambiguous outside those conditions.

The results use the main paper's finite-horizon unified action value, including cost, positive duration, operation during research, the joint completion law, compressed updating, and lower-horizon optimized continuation. They add no path–outcome independence assumption.

S2.1 Closure increments and relative saturation

Definition S2.1 (Closure increment and interaction)**.** For $F_0 \le F_1$ and $C_0 \subseteq C_1$, define

$$\Delta_C V_h(F; C_1, C_0) := V_h(F, C_1) - V_h(F, C_0)$$

and

$$J_h := \Delta_C V_h(F_1; C_1, C_0) - \Delta_C V_h(F_0; C_1, C_0).$$

The two margins are substitutes on the rectangle when $J_h \le 0$ and complements when $J_h \ge 0$.

Frontier independence holds project availability, initiation cost, duration, operation during research, and the complete joint completion law fixed across the two frontiers. Closure expansion makes every closure-poor action available under the rich closure. These restrictions make each fixed opportunity saturate, but they do not compare saturation across projects. The theorem therefore needs a relative condition at the action level.

Theorem S2.2 (Frontier–closure substitution under relative saturation)**.** *Consider the four realizable compressed states*

$$(F_0, C_0), \quad (F_0, C_1), \quad (F_1, C_0), \quad (F_1, C_1), \qquad F_0 \le F_1, \quad C_0 \subseteq C_1.$$

Suppose research primitives are frontier independent and the closure-rich menu contains the closure-poor menu. Suppose also that, at every finite Bellman node, every action a_1 feasible with C_1 and every action a_0 feasible with C_0 satisfy

$$Q_h(F_1, C_1, a_1) - Q_h(F_1, C_0, a_0) \le Q_h(F_0, C_1, a_1) - Q_h(F_0, C_0, a_0). \tag{S2.1}$$

Here Q_h is the unified action value, including cost, positive duration, operation during research, the joint completion law, compressed updating, and lower-horizon optimized continuation. Then, for every finite horizon and belief,

$$\Delta_C V_h(F_1; C_1, C_0) \le \Delta_C V_h(F_0; C_1, C_0), \qquad \textit{equivalently } J_h \le 0.$$

At a fixed finite horizon and belief, for the four realizable corners choose a_{11} maximizing at (F_1, C_1) and a_{00} maximizing at (F_0, C_0). Finiteness of the action menus supplies these maximizers. Frontier independence transports them to the comparison corners; no probabilistic independence is asserted. Relative action saturation and finite maximum attainment give

$$\begin{aligned} V_h(F_1, C_1) - V_h(F_1, C_0) &\le Q_h(F_1, C_1, a_{11}) - Q_h(F_1, C_0, a_{00}) \\ &\le Q_h(F_0, C_1, a_{11}) - Q_h(F_0, C_0, a_{00}) \\ &\le V_h(F_0, C_1) - V_h(F_0, C_0). \end{aligned}$$

This is exactly $J_h \le 0$. Neither detectability nor contraction is used. Without the middle inequality, individual project saturation does not control optimizer switching.

S2.2 Primitive common-gap sufficient conditions

Proposition S2.3 (Primitive common-gap frontier saturation)**.** *Maintain the realizable rectangle, frontier-independent primitives, and menu inclusion of Theorem S2.2. Suppose that at every finite Bellman node and belief the four action-value tables admit*

$$\begin{aligned} Q_h(F_i, C_1, a) &= B_{h,i} + \eta^1_{h,a} + \lambda^1_{h,a} g_{h,i}, \\ Q_h(F_i, C_0, a) &= B_{h,i} + \eta^0_{h,a} + \lambda^0_{h,a} g_{h,i}, \qquad i \in \{0, 1\}, \end{aligned} \tag{S2.2}$$

where $g_{h,1} \le g_{h,0}$, every rich-menu exposure satisfies $\lambda^1_{h,a} \ge 0$, and every feasible poor-menu action satisfies $\lambda^0_{h,a} = 0$. Then relative action saturation (S2.1) *holds. Consequently*

$$\Delta_C V_h(F_1; C_1, C_0) \le \Delta_C V_h(F_0; C_1, C_0)$$

for every finite horizon and belief.

The subclass is economically narrow but primitive. A fixed descendant profile supplies the common positive-part gap g; frontier-independent cost, duration, operation, candidate, admission, and fixed-continuation terms enter the intercepts and nonnegative exposures. The poor closure is Continue-only in the leading case; more generally its feasible actions may have fixed payoffs but no exposure to the descendant gap. The shared decomposition across frontiers is the required fixed-continuation factorization. It is an assumption defining this subclass, not a claim that arbitrary unified continuation preserves the form.

There is a transition-level sufficient condition for the displayed gap order. Let the low- and high-frontier gaps share the process belief kernel P and discount $\beta \ge 0$, and let

$$\begin{aligned} g_{0,i}(b) &= \bar{g}_i(b), \\ g_{h+1,i}(b) &= s_i(b) + \beta \sum_{b'} P(b, b') g_{h,i}(b'), \qquad i \in \{0, 1\}. \end{aligned}$$

If $\bar{g}_1 \le \bar{g}_0$ and $s_1 \le s_0$ pointwise, positivity of the fixed transition operator gives $g_{h,1} \le g_{h,0}$ by finite-horizon induction. Thus the common-gap proposition is recursion stable when the descendant continuation uses this fixed kernel and remains outside the action-dependent optimizer.

The induction makes this preservation step explicit. Assume $\beta \ge 0$, $P(b, b') \ge 0$, $\bar{g}_1(b) \le \bar{g}_0(b)$, and $s_1(b) \le s_0(b)$ pointwise. The terminal order is the induction base. If $g_{h,1}(b') \le g_{h,0}(b')$ for every b', then

$$\begin{aligned} g_{h+1,1}(b) - g_{h+1,0}(b) &= s_1(b) - s_0(b) \\ &\quad + \beta \sum_{b'} P(b, b') \big(g_{h,1}(b') - g_{h,0}(b')\big) \le 0. \end{aligned}$$

Hence $g_{h,1} \le g_{h,0}$ for every finite h. Row mass one gives the kernel its probability interpretation; positivity, rather than normalization, is the sign-bearing property in the induction.

At each node, fixed project and continuation primitives give

$$Q_h(F_i, C_j, a) = B_{h,i} + \eta^j_{h,a} + \lambda^j_{h,a} g_{h,i}.$$

Take a rich action a_1 and a feasible poor action a_0. The frontier-specific base cancels within their relative payoff. Because every poor action has $\lambda^0_{h,a_0} = 0$, its change from the low to the high frontier is

$$\begin{aligned} &\big[Q_h(F_1, C_1, a_1) - Q_h(F_1, C_0, a_0)\big] \\ &\quad - \big[Q_h(F_0, C_1, a_1) - Q_h(F_0, C_0, a_0)\big] = \lambda^1_{h,a_1}(g_{h,1} - g_{h,0}) \le 0, \end{aligned}$$

using $\lambda^1_{h,a_1} \ge 0$. This is precisely relative action saturation. Frontier independence transports the high-rich and low-poor maximizers to the other corners, and the finite-max comparison above then yields $J_h \le 0$.

S2.3 Complementarity and sign-ambiguous cases

The zero-exposure restriction cannot be replaced by merely ordering the added project's exposure above the incumbent project's. Because Continue is always available and has zero exposure, its advantage over a positive-exposure poor project rises when the common gap shrinks. An exact 2,430-row rational search finds 648 failures of all-pairs relative saturation under the broader exposure order and none among the 810 zero-poor-exposure rows. This is finite falsification and validation, not proof; the proposition is separately kernel checked.

The substitution theorem is sharp. Exact rational finite examples attain all three interaction signs:

construction	$\Delta_C V(F_0)$	$\Delta_C V(F_1)$	J
fixed opportunity with saturation	2	1	-1
frontier-dependent success $0 \to 1$	0	1/2	1/2
separable closure bonus	3/2	3/2	0

Proposition S2.4 (Complementarity from project switching)**.** *Frontier independence and individual project saturation do not imply substitution. Let $\beta = 1/2$, the descendant payoff be* 10*, and $(F_0, F_1) = (0, 8)$. The old project has fixed $(\pi, \kappa) = (1, 2)$, while closure adds a project with fixed $(\pi, \kappa) = (1/2, 0)$. Their premia fall respectively*

$$3 \to 0 \qquad \textit{and} \qquad 5/2 \to 1/2.$$

Nevertheless, optimization gives

$$\Delta_C V(F_0) = 0, \qquad \Delta_C V(F_1) = 1/2, \qquad J = 1/2.$$

The added project's advantage over the old project rises from $-1/2$ to $1/2$, violating relative action saturation. This produces strict complementarity even though both project premia decline individually. The example remains outside Proposition S2.3: its poor-menu project has positive common-gap exposure. Frontier-dependent success provides a second exact complementarity mechanism; the fixed-opportunity and separable examples show that substitution and zero interaction are also attainable. Hence neither closure enrichment nor frontier improvement has an unconditional interaction sign.

The numerical interaction response remains evidence separate from the theory. It preserves the registered 2,430-row frontier–project boundary grid and five canonical four-corner fixtures: primitive strict substitution, the saturation boundary, optimizer-switching complementarity, frontier-dependent-success complementarity, and nonconstant separability. Every row reports the two closure increments, their cross difference, the four selected projects, four realizability flags, and the primitive sufficient-condition certificate. Only jointly realizable four-corner rows enter any sign frequency; nonrealizable diagnostic rows remain recorded but are not aggregated.

S3 Additional Comparative Statics and Counterexamples

This section records the supporting one-way comparative statics and the exact counterexamples that delimit the coverage results in Section S1. Every comparison holds conditional on the displayed current state and fixed action or on the stated one-shot problem. Unnamed primitives are held fixed. These results do not promote a one-shot cost-covering region to an optimized Bellman research region.

S3.1 Cost, admission, survival, and delay

The supporting order comparisons are compact because every nondisplayed primitive is held fixed:

Change	Direction	Required boundary
higher initiation cost	optimized value weakly falls	the same project actions change; Continue and other primitives are fixed
higher admission or survival	project value weakly rises	successful completion weakly dominates failure
longer duration	project value weakly falls	every added operating date satisfies $F_t \le (1-\beta)W$; suspended operation is $F_t = 0$
higher incumbent frontier	fixed-candidate operating gain weakly falls	candidate profile and research opportunities are fixed

Proposition S3.1 (One-shot cutoff comparative statics)**.** *Holding all other one-shot primitives fixed, higher cost shrinks $\mathcal{C}$; higher nonnegative survival or admission expands it; and a higher operating frontier shrinks it for a fixed candidate. When both compared nonempty regions have upper-threshold representations, higher cost or frontier raises the cutoff, while higher survival or admission lowers it.*

These set inclusions concern the one-shot cost-covering region in Definition S1.9. Cost and survival also order an exact finite weak-research region for a fixed research action versus Continue when all other unified-model returns are unchanged. Neither result, by itself, orders the full optimal research region when opportunity menus, continuation laws, or competing actions also change.

The cutoff directions are pointwise set comparisons. Raising κ removes weakly covered beliefs. Raising a nonnegative survival or admission factor raises G and adds beliefs. Raising the frontier lowers the positive-part gap and removes beliefs. If two nonempty regions are $\mathrm{Ici}(b_0^*)$ and $\mathrm{Ici}(b_1^*)$, set inclusion reverses the cutoff order, yielding the directions in Proposition S3.1.

At a fixed finite horizon and conditional on the current decision state, holding every nondisplayed primitive fixed, a higher initiation cost subtracts the same nonnegative increment from each affected project action and leaves Continue unchanged. Backward induction and maximization over the finite action menu therefore make optimized value antitone in the cost schedule.

For a binary completion law, condition on the fixed current state and chosen project, with success continuation W^+ and failure continuation W^-. The binary specialization assumes the conditional success mass $m = \pi\rho^d$; it is not inferred from unrelated marginals. The completion component is

$$mW^+ + (1-m)W^-.$$

For two admission–survival configurations its exact difference is

$$\left[m_1 W^+ + (1-m_1)W^-\right] - \left[m_0 W^+ + (1-m_0)W^-\right] = (m_1 - m_0)(W^+ - W^-).$$

Thus greater admission or survival raises project value when $W^+ \ge W^-$, and reverses sign if success is harmful.

For continued operation, keep the initial state, belief path, and terminal continuation W fixed while comparing durations. Write the elapsed return at duration d as

$$R_d = -\kappa + \sum_{t<d} \beta^t F_t + \beta^d W.$$

The one-date increment is

$$R_{d+1} - R_d = \beta^d \left[F_d - (1-\beta)W\right].$$

Consequently, for $d_0 < d_1$,

$$R_{d_1} - R_{d_0} = \sum_{t=d_0}^{d_1-1} \beta^t [F_t - (1-\beta)W] .$$

Longer delay is therefore harmful when every added date satisfies $F_t \le (1-\beta)W$. Suspended operation is the special case $F_t = 0$. Nonnegativity alone is insufficient: with $\beta = 1/2$ and $F_t = W = 1$, $R_1 = 3/2 < R_2 = 7/4$.

S3.2 Threshold and multi-gap counterexamples

The assumptions of the positive threshold theorem have exact finite boundaries. The following examples preserve the original rational values and isolate the failed condition in each case.

Counterexample S3.2 (A nonmonotone kernel defeats the upper threshold)**.** On $B = \{0, 1, 2\}$, take deterministic destinations $(1, 0, 1)$. The resulting row-stochastic kernel maps the increasing nonnegative gap $\Delta = (0, 1, 2)$ to $P\Delta = (1, 0, 1)$. With unit survival and antitone constant unit cost, the weak cost-covering set is $\{0, 2\}$, which is neither upper nor order-connected. Thus the stochastic-monotonicity assumption is independently necessary. The same kernel also maps the single-peaked gap $(0, 1, 0)$ to $(1, 0, 1)$, retaining the earlier boundary against unrestricted single-peaked preservation.

Counterexample S3.3 (Non-antitone cost disconnects a monotone covering set)**.** The increasing potential $(1, 2, 3)$ and cost $(0, 3, 0)$ produce the cost-covering set $\{1, 3\}$. Antitonicity of research cost cannot be dropped from Theorem S1.10.

Counterexample S3.4 (One-project multi-gap disconnection)**.** Let $B = \{0, 1, 2, 3, 4\}$. One project supplies candidates with certified gaps

$$\Delta^L = (4, 0, 0, 0, 0), \qquad \Delta^R = (0, 0, 0, 0, 4)$$

over the zero frontier. Under the degree-four Bernstein/binomial kernel

$$P_{ij} = \binom{4}{j} (i/4)^j (1 - i/4)^{4-j}, \qquad i, j \in B,$$

their aggregate one-shot coverage potential is $P(\Delta^L + \Delta^R) = (4, 41/32, 1/2, 41/32, 4)$. Constant strict cost one therefore gives the one-shot cost-covering set $\{0, 1\} \cup \{3, 4\}$, which is not order-connected.

Counterexample S3.5 (Arbitrary cost defeats a component bound)**.** With the same five-state kernel, constant gap and potential $(2, 2, 2, 2, 2)$, and cost $(0, 3, 0, 3, 0)$, the strict cost-covering set is $\{0, 2, 4\}$. A kernel-only variation-diminishing condition cannot control net-region components under unrestricted belief-dependent cost.

The multi-gap example shows that the upper-threshold conclusion does not extend to separated candidate gaps. Separately, the first two examples show that neither monotone gaps alone nor a well-behaved gross-value profile alone ensures connected cost covering. The arbitrary-cost example rules out a kernel-only component bound when belief-dependent cost is unrestricted.

S3.3 Auxiliary finite-model illustration

The following exact construction isolates the distinction between operating profiles and generative modules in the main paper's finite model. It is an illustration rather than an additional assumption or theorem.

Example S3.6 (A two-module generator)**.** Let $X = \{\text{down}, \text{up}\}$, let $\mu_{b^-} = (3/4, 1/4)$ and $\mu_{b^+} = (1/4, 3/4)$, and take identity closure. Strategies s_T and s_H carry modules m_T and m_H. A project q requires both. When $\{m_T, m_H\} \subseteq C$, set

$$G(q,b,C)(\text{some}(c)) = \frac{2}{3}, \qquad G(q,b,C)(\text{none}) = \frac{1}{3}, \qquad \nu(q,b,C,c) = \frac{3}{4};$$

otherwise set $G(q,b,C) = \delta_{\text{none}}$. Retaining both modules therefore admits c with probability 1/2, while deleting the sole carrier of either module makes admission impossible. If $u_c(\text{down}) = -1$ and $u_c(\text{up}) = 2$, then $j_c(b^-) = -1/4$ and $j_c(b^+) = 5/4$: current operating payoff depends on the filtered state, while candidate availability depends on retained closure. For a positive-duration instance, take $d_q = 2$, initiation cost 1/10, and operation flag $o_q = 1$.

S3.4 Conditional replacement under a capacity bound

The supporting question is what an incumbent should release when an outer-certified candidate $c \notin L$ becomes available for retention and capacity binds. This formulation is conditional on outer certification and is separate from the raw generation, verification, and admission law in the main paper.

Let the incumbent satisfy $W(L) \le B$, let $w_c \le B$, and permit deletion of active incumbents $D \subseteq L \setminus \{s_0\}$. Additivity makes the required release

$$r_c(B) := [W(L) + w_c - B]_+,$$

so the feasible deletion family is

$$\mathcal{D}_c(B) := \{D \subseteq L \setminus \{s_0\} : W(D) \ge r_c(B)\}.$$

The conditional replacement problem is

$$\underset{D \in \mathcal{D}_c(B)}{\text{maximize}}\, V_\theta(b, (L \setminus D) \cup \{c\}). \tag{S3.1}$$

For interpretation, define the unconstrained candidate gain and displacement loss

$$\begin{aligned} G_c(b, L) &:= V_\theta(b, L \cup \{c\}) - V_\theta(b, L), \\ \ell_c(D) &:= V_\theta(b, L \cup \{c\}) - V_\theta(b, (L \setminus D) \cup \{c\}). \end{aligned}$$

Then the best net admission value is the definition-level identity

$$A_c(b, L, B) = G_c(b, L) - \min_{D \in \mathcal{D}_c(B)} \ell_c(D).$$

Under common insertion and the productive factorization, a deletion safe before admission is sufficient for zero displacement loss, but it is not necessary: the candidate may replace a capability carried by a deleted incumbent. This supporting formulation is not part of the Lean-verified theorem set, and it makes no constrained–penalized equivalence claim.

S4 Complete Canonical Numerical Records

This section collects the full numerical records referenced by the main paper's Appendix D. The tables below are typeset directly from the committed CSV files; compilation does not rerun either numerical solver. Exact rational fields retain their machine-readable `numerator//denominator` form. The compact stationary table is duplicated from the main paper to make the record self-contained.

S4.1 Convergence history and stationary solution

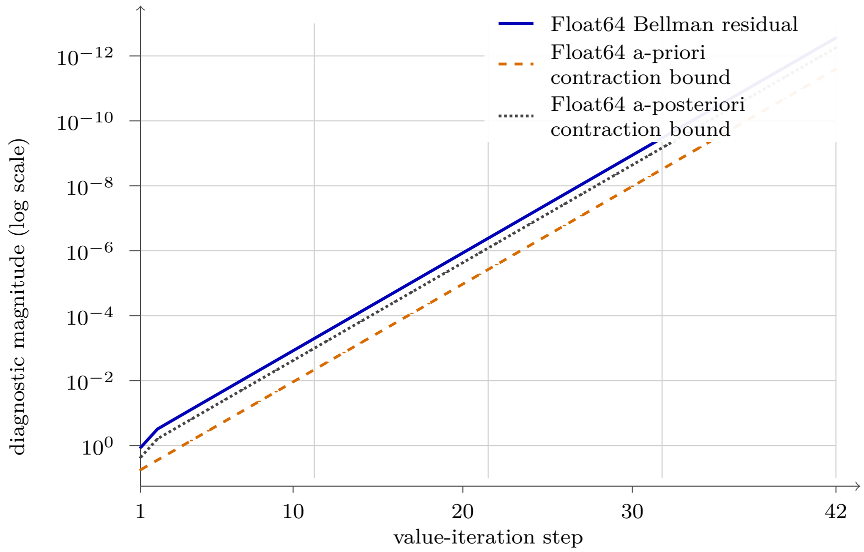

Figure 1: Complete Float64 convergence history for the unified six-state benchmark. Markers are the 42 committed iterations; joining segments are visual guides. The dotted series is the Float64 a-posteriori contraction bound. Exact rational policy iteration is a separate calculation with Bellman residual zero.

The final convergence-row bound is $r^{64}_{42}/(1-\beta_{64}) = 5.54223 \times 10^{-13}$, computed in Float64. The canonical summary separately converts the binary iterate exactly to `Rational{BigInt}`, records maximum exact error $937/1688849860263936 \approx 5.5482 \times 10^{-13}$, and evaluates the exact rational Bellman residual to obtain the exactly re-evaluated residual-based certificate $2499/4503599627370496 \approx 5.5489 \times 10^{-13}$. The latter is not the residual-bound column in the convergence table.

Table 1: Full 42-row Float64 convergence record.

Iter.	Float64 increment	Float64 Bellman residual	Float64 a-priori contraction bound	Float64 a-posteriori contraction bound
1	5.5625	1.15625	5.5625	2.3125
2	1.15625	0.30859375	2.78125	0.6171875
3	0.30859375	0.1533203125	1.390625	0.306640625
4	0.1533203125	0.076416015625	0.6953125	0.15283203125
5	0.076416015625	0.03814697265625	0.34765625	0.0762939453125
6	0.03814697265625	0.0190582275390625	0.173828125	0.038116455078125
7	0.0190582275390625	0.009525299072265625	0.0869140625	0.01905059814453125
8	0.009525299072265625	0.0047616958618164062	0.04345703125	0.0095233917236328125
9	0.0047616958618164062	0.0023806095123291016	0.021728515625	0.0047612190246582031
10	0.0023806095123291016	0.0011902451515197754	0.0108642578125	0.0023804903030395508
11	0.0011902451515197754	0.00059510767459869385	0.00543212890625	0.0011902153491973877
12	0.00059510767459869385	0.00029755011200904846	0.002716064453125	0.00059510022401809692
13	0.00029755011200904846	0.00014877412468194962	0.0013580322265625	0.00029754824936389923
14	0.00014877412468194962	7.4386829510331154e-05	0.00067901611328125	0.00014877365902066231
15	7.4386829510331154e-05	3.7193356547504663e-05	0.000339508056640625	7.4386713095009327e-05
16	3.7193356547504663e-05	1.8596663721837103e-05	0.0001697540283203125	3.7193327443674207e-05
17	1.8596663721837103e-05	9.2983282229397446e-06	8.487701416015625e-05	1.8596656445879489e-05
18	9.2983282229397446e-06	4.6491632019751705e-06	4.2438507080078125e-05	9.298326403950341e-06
19	4.6491632019751705e-06	2.3245813736139098e-06	2.1219253540039062e-05	4.6491627472278196e-06
20	2.3245813736139098e-06	1.162290629963536e-06	1.0609626770019531e-05	2.3245812599270721e-06
21	1.162290629963536e-06	5.8114530077091331e-07	5.3048133850097656e-06	1.1622906015418266e-06
22	5.8114530077091331e-07	2.9057264683274298e-07	2.6524066925048828e-06	5.8114529366548595e-07
23	2.9057264683274298e-07	1.4528632252819307e-07	1.3262033462524414e-06	2.9057264505638614e-07
24	1.4528632252819307e-07	7.2643161708185744e-08	6.631016731262207e-07	1.4528632341637149e-07
25	7.2643161708185744e-08	3.6321580410003662e-08	3.3155083656311035e-07	7.2643160820007324e-08
26	3.6321580410003662e-08	1.8160790205001831e-08	1.6577541828155518e-07	3.6321580410003662e-08
27	1.8160790205001831e-08	9.0803951025009155e-09	8.2887709140777588e-08	1.8160790205001831e-08
28	9.0803951025009155e-09	4.5401975512504578e-09	4.1443854570388794e-08	9.0803951025009155e-09
29	4.5401975512504578e-09	2.2700987756252289e-09	2.0721927285194397e-08	4.5401975512504578e-09
30	2.2700987756252289e-09	1.1350493878126144e-09	1.0360963642597198e-08	2.2700987756252289e-09
31	1.1350493878126144e-09	5.6752469390630722e-10	5.1804818212985992e-09	1.1350493878126144e-09
32	5.6752469390630722e-10	2.8376234695315361e-10	2.5902409106492996e-09	5.6752469390630722e-10
33	2.8376234695315361e-10	1.4188117347657681e-10	1.2951204553246498e-09	2.8376234695315361e-10
34	1.4188117347657681e-10	7.0940586738288403e-11	6.4756022766232491e-10	1.4188117347657681e-10
35	7.0940586738288403e-11	3.5470293369144201e-11	3.2378011383116245e-10	7.0940586738288403e-11
36	3.5470293369144201e-11	1.7735146684572101e-11	1.6189005691558123e-10	3.5470293369144201e-11
37	1.7735146684572101e-11	8.8675733422860503e-12	8.0945028457790613e-11	1.7735146684572101e-11
38	8.8675733422860503e-12	4.4337866711430252e-12	4.0472514228895307e-11	8.8675733422860503e-12
39	4.4337866711430252e-12	2.2168933355715126e-12	2.0236257114447653e-11	4.4337866711430252e-12
40	2.2168933355715126e-12	1.1084466677857563e-12	1.0118128557223827e-11	2.2168933355715126e-12
41	1.1084466677857563e-12	5.5422333389287814e-13	5.0590642786119133e-12	1.1084466677857563e-12
42	5.5422333389287814e-13	2.7711166694643907e-13	2.5295321393059567e-12	5.5422333389287814e-13

State	Belief	$V^\star$ (payoff units)	$\pi^\star$	Winning margin (payoff units)
K_0	ℓ	$\frac{115}{288}$	Discover	$\frac{23}{96}$
K_0	h	$\frac{23}{288}$	Continue	$\frac{245}{384}$
K_1	ℓ	1	Scale	$\frac{23}{48}$
K_1	h	$\frac{7}{6}$	Scale	$\frac{29}{48}$
K_2	ℓ	$\frac{14}{3}$	Continue	$\frac{3}{16}$
K_2	h	$\frac{22}{3}$	Continue	$\frac{3}{16}$

Table 2: Exact rational stationary solution, duplicated from the main paper.

Table 3: Complete stationary action-value record.

State	Belief	Value	Chosen	Second	Continue (Q)	Research	Research (Q)	Margin	Duration	Reward block	Discounted continuation
K0	low	115//288	research:discover	continue	23//144	research:discover	115//288	23//96	1	-1//32	31//72
K0	high	23//288	continue	research:discover	23//288	research:discover	-643//1152	245//384	1	-1//1	509//1152
K1	low	1//1	research:scale	continue	25//48	research:scale	1//1	23//48	2	-3//16	19//16
K1	high	7//6	research:scale	continue	9//16	research:scale	7//6	29//48	2	-3//16	65//48
K2	low	14//3	continue	research:scale	14//3	research:scale	215//48	3//16	2	49//16	17//12
K2	high	22//3	continue	research:scale	22//3	research:scale	343//48	3//16	2	89//16	19//12

S4.2 Horizon-by-horizon values and actions

The value record includes horizons zero through eight and all four stationary solution/evaluation outputs. The policy record includes every positive finite horizon and both stationary policy outputs. This retains the exact and floating-point rows together rather than presenting a selectively rounded comparison.

Table 4: Complete horizon-by-horizon and stationary value record.

Solution	Horizon	Arithmetic	State	Belief	Value
finite_horizon_exact	0	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	0	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	0	Rational{BigInt}	K1	low	0//1
finite_horizon_exact	0	Rational{BigInt}	K1	high	0//1
finite_horizon_exact	0	Rational{BigInt}	K2	low	0//1
finite_horizon_exact	0	Rational{BigInt}	K2	high	0//1
finite_horizon_exact	1	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	1	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	1	Rational{BigInt}	K1	low	0//1
finite_horizon_exact	1	Rational{BigInt}	K1	high	0//1
finite_horizon_exact	1	Rational{BigInt}	K2	low	2//1
finite_horizon_exact	1	Rational{BigInt}	K2	high	4//1
finite_horizon_exact	2	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	2	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	2	Rational{BigInt}	K1	low	0//1
finite_horizon_exact	2	Rational{BigInt}	K1	high	0//1
finite_horizon_exact	2	Rational{BigInt}	K2	low	13//4
finite_horizon_exact	2	Rational{BigInt}	K2	high	23//4
finite_horizon_exact	3	Rational{BigInt}	K0	low	0//1
finite_horizon_exact	3	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	3	Rational{BigInt}	K1	low	3//8
finite_horizon_exact	3	Rational{BigInt}	K1	high	1//2
finite_horizon_exact	3	Rational{BigInt}	K2	low	63//16
finite_horizon_exact	3	Rational{BigInt}	K2	high	105//16
finite_horizon_exact	4	Rational{BigInt}	K0	low	31//256
finite_horizon_exact	4	Rational{BigInt}	K0	high	0//1
finite_horizon_exact	4	Rational{BigInt}	K1	low	21//32
finite_horizon_exact	4	Rational{BigInt}	K1	high	13//16
finite_horizon_exact	4	Rational{BigInt}	K2	low	275//64
finite_horizon_exact	4	Rational{BigInt}	K2	high	445//64
finite_horizon_exact	5	Rational{BigInt}	K0	low	1973//8192
finite_horizon_exact	5	Rational{BigInt}	K0	high	31//2048
finite_horizon_exact	5	Rational{BigInt}	K1	low	105//128
finite_horizon_exact	5	Rational{BigInt}	K1	high	63//64
finite_horizon_exact	5	Rational{BigInt}	K2	low	1147//256
finite_horizon_exact	5	Rational{BigInt}	K2	high	1829//256
finite_horizon_exact	6	Rational{BigInt}	K0	low	82523//262144

Solution	Horizon	Arithmetic	State	Belief	Value
finite_horizon_exact	6	Rational{BigInt}	K0	high	2345//65536
finite_horizon_exact	6	Rational{BigInt}	K1	low	465//512
finite_horizon_exact	6	Rational{BigInt}	K1	high	275//256
finite_horizon_exact	6	Rational{BigInt}	K2	low	4683//1024
finite_horizon_exact	6	Rational{BigInt}	K2	high	7413//1024
finite_horizon_exact	7	Rational{BigInt}	K0	low	2982325//8388608
finite_horizon_exact	7	Rational{BigInt}	K0	high	110663//2097152
finite_horizon_exact	7	Rational{BigInt}	K1	low	1953//2048
finite_horizon_exact	7	Rational{BigInt}	K1	high	1147//1024
finite_horizon_exact	7	Rational{BigInt}	K2	low	18923//4096
finite_horizon_exact	7	Rational{BigInt}	K2	high	29845//4096
finite_horizon_exact	8	Rational{BigInt}	K0	low	101185083//268435456
finite_horizon_exact	8	Rational{BigInt}	K0	high	4310281//67108864
finite_horizon_exact	8	Rational{BigInt}	K1	low	8001//8192
finite_horizon_exact	8	Rational{BigInt}	K1	high	4683//4096
finite_horizon_exact	8	Rational{BigInt}	K2	low	76075//16384
finite_horizon_exact	8	Rational{BigInt}	K2	high	119765//16384
stationary_exact_policy_iteration		Rational{BigInt}	K0	low	115//288
stationary_exact_policy_evaluation		Rational{BigInt}	K0	low	115//288
stationary_float64_value_iteration		Float64	K0	low	0.39930555555531799
stationary_float64_policy_evaluation		Float64	K0	low	0.39930555555555552
stationary_exact_policy_iteration		Rational{BigInt}	K0	high	23//288
stationary_exact_policy_evaluation		Rational{BigInt}	K0	high	23//288
stationary_float64_value_iteration		Float64	K0	high	0.079861111110873587
stationary_float64_policy_evaluation		Float64	K0	high	0.079861111111111105
stationary_exact_policy_iteration		Rational{BigInt}	K1	low	1//1
stationary_exact_policy_evaluation		Rational{BigInt}	K1	low	1//1
stationary_float64_value_iteration		Float64	K1	low	0.99999999999976241
stationary_float64_policy_evaluation		Float64	K1	low	1
stationary_exact_policy_iteration		Rational{BigInt}	K1	high	7//6
stationary_exact_policy_evaluation		Rational{BigInt}	K1	high	7//6
stationary_float64_value_iteration		Float64	K1	high	1.1666666666664289
stationary_float64_policy_evaluation		Float64	K1	high	1.1666666666666667
stationary_exact_policy_iteration		Rational{BigInt}	K2	low	14//3
stationary_exact_policy_evaluation		Rational{BigInt}	K2	low	14//3
stationary_float64_value_iteration		Float64	K2	low	4.6666666666661119
stationary_float64_policy_evaluation		Float64	K2	low	4.666666666666667
stationary_exact_policy_iteration		Rational{BigInt}	K2	high	22//3
stationary_exact_policy_evaluation		Rational{BigInt}	K2	high	22//3
stationary_float64_value_iteration		Float64	K2	high	7.3333333333327788
stationary_float64_policy_evaluation		Float64	K2	high	7.3333333333333339

Table 5: Complete horizon-by-horizon and stationary action record.

Solution	Horizon	State	Belief	Action
finite_horizon_exact	1	K0	low	continue
finite_horizon_exact	1	K0	high	continue
finite_horizon_exact	1	K1	low	continue
finite_horizon_exact	1	K1	high	continue
finite_horizon_exact	1	K2	low	continue
finite_horizon_exact	1	K2	high	continue
finite_horizon_exact	2	K0	low	continue
finite_horizon_exact	2	K0	high	continue
finite_horizon_exact	2	K1	low	continue
finite_horizon_exact	2	K1	high	continue
finite_horizon_exact	2	K2	low	continue
finite_horizon_exact	2	K2	high	continue
finite_horizon_exact	3	K0	low	continue
finite_horizon_exact	3	K0	high	continue
finite_horizon_exact	3	K1	low	research:scale
finite_horizon_exact	3	K1	high	research:scale
finite_horizon_exact	3	K2	low	continue
finite_horizon_exact	3	K2	high	continue
finite_horizon_exact	4	K0	low	research:discover
finite_horizon_exact	4	K0	high	continue
finite_horizon_exact	4	K1	low	research:scale
finite_horizon_exact	4	K1	high	research:scale
finite_horizon_exact	4	K2	low	continue
finite_horizon_exact	4	K2	high	continue
finite_horizon_exact	5	K0	low	research:discover
finite_horizon_exact	5	K0	high	continue
finite_horizon_exact	5	K1	low	research:scale
finite_horizon_exact	5	K1	high	research:scale
finite_horizon_exact	5	K2	low	continue
finite_horizon_exact	5	K2	high	continue
finite_horizon_exact	6	K0	low	research:discover
finite_horizon_exact	6	K0	high	continue
finite_horizon_exact	6	K1	low	research:scale
finite_horizon_exact	6	K1	high	research:scale
finite_horizon_exact	6	K2	low	continue
finite_horizon_exact	6	K2	high	continue
finite_horizon_exact	7	K0	low	research:discover
finite_horizon_exact	7	K0	high	continue
finite_horizon_exact	7	K1	low	research:scale
finite_horizon_exact	7	K1	high	research:scale
finite_horizon_exact	7	K2	low	continue
finite_horizon_exact	7	K2	high	continue
finite_horizon_exact	8	K0	low	research:discover
finite_horizon_exact	8	K0	high	continue
finite_horizon_exact	8	K1	low	research:scale

Solution	Horizon	State	Belief	Action
`finite_horizon_exact`	8	K1	high	`research:scale`
`finite_horizon_exact`	8	K2	low	`continue`
`finite_horizon_exact`	8	K2	high	`continue`
`stationary_exact_policy_iteration`		K0	low	`research:discover`
`stationary_float64_value_iteration`		K0	low	`research:discover`
`stationary_exact_policy_iteration`		K0	high	`continue`
`stationary_float64_value_iteration`		K0	high	`continue`
`stationary_exact_policy_iteration`		K1	low	`research:scale`
`stationary_float64_value_iteration`		K1	low	`research:scale`
`stationary_exact_policy_iteration`		K1	high	`research:scale`
`stationary_float64_value_iteration`		K1	high	`research:scale`
`stationary_exact_policy_iteration`		K2	low	`continue`
`stationary_float64_value_iteration`		K2	low	`continue`
`stationary_exact_policy_iteration`		K2	high	`continue`
`stationary_float64_value_iteration`		K2	high	`continue`

S4.3 Path, admission, and operating-reward atoms

Table 6: Complete positive-duration path and admission-atom record.

Project	Start state	Start belief	Duration	Path	Terminal belief	Admitted	Joint mass	Path mass	Conditional admission
discover	K0	low	1	low>low	low	failure	3//16	3//4	1//4
discover	K0	low	1	low>low	low	carrier_a	3//8	3//4	1//2
discover	K0	low	1	low>low	low	carrier_b	3//16	3//4	1//4
discover	K0	low	1	low>high	high	failure	1//16	1//4	1//4
discover	K0	low	1	low>high	high	carrier_a	1//8	1//4	1//2
discover	K0	low	1	low>high	high	carrier_b	1//16	1//4	1//4
discover	K0	high	1	high>low	low	failure	1//16	1//4	1//4
discover	K0	high	1	high>low	low	carrier_a	1//8	1//4	1//2
discover	K0	high	1	high>low	low	carrier_b	1//16	1//4	1//4
discover	K0	high	1	high>high	high	failure	3//16	3//4	1//4
discover	K0	high	1	high>high	high	carrier_a	3//8	3//4	1//2
discover	K0	high	1	high>high	high	carrier_b	3//16	3//4	1//4
scale	K1	low	2	low>low>low	low	failure	9//40	9//16	2//5
scale	K1	low	2	low>low>low	low	descendant	27//80	9//16	3//5
scale	K1	low	2	low>low>high	high	descendant	3//16	3//16	1//1
scale	K1	low	2	low>high>low	low	failure	1//40	1//16	2//5
scale	K1	low	2	low>high>low	low	descendant	3//80	1//16	3//5
scale	K1	low	2	low>high>high	high	descendant	3//16	3//16	1//1
scale	K1	high	2	high>low>low	low	failure	1//8	3//16	2//3
scale	K1	high	2	high>low>low	low	descendant	1//16	3//16	1//3
scale	K1	high	2	high>low>high	high	descendant	1//16	1//16	1//1
scale	K1	high	2	high>high>low	low	failure	1//8	3//16	2//3
scale	K1	high	2	high>high>low	low	descendant	1//16	3//16	1//3
scale	K1	high	2	high>high>high	high	descendant	9//16	9//16	1//1
scale	K2	low	2	low>low>low	low	failure	9//40	9//16	2//5
scale	K2	low	2	low>low>low	low	descendant	27//80	9//16	3//5
scale	K2	low	2	low>low>high	high	descendant	3//16	3//16	1//1
scale	K2	low	2	low>high>low	low	failure	1//40	1//16	2//5
scale	K2	low	2	low>high>low	low	descendant	3//80	1//16	3//5
scale	K2	low	2	low>high>high	high	descendant	3//16	3//16	1//1
scale	K2	high	2	high>low>low	low	failure	1//8	3//16	2//3
scale	K2	high	2	high>low>low	low	descendant	1//16	3//16	1//3
scale	K2	high	2	high>low>high	high	descendant	1//16	1//16	1//1
scale	K2	high	2	high>high>low	low	failure	1//8	3//16	2//3
scale	K2	high	2	high>high>low	low	descendant	1//16	3//16	1//3
scale	K2	high	2	high>high>high	high	descendant	9//16	9//16	1//1

Table 7: Complete operating-reward block record.

State	Belief	Action	Duration	Operates	Initiation cost	Incumbent reward	Net reward block
K0	low	continue	1	false	0//1	0//1	0//1
K0	low	discover	1	true	1//32	0//1	-1//32
K0	high	continue	1	false	0//1	0//1	0//1
K0	high	discover	1	true	1//1	0//1	-1//1
K1	low	continue	1	false	0//1	0//1	0//1
K1	low	scale	2	true	3//16	0//1	-3//16
K1	high	continue	1	false	0//1	0//1	0//1
K1	high	scale	2	true	3//16	0//1	-3//16
K2	low	continue	1	false	0//1	2//1	2//1
K2	low	scale	2	true	3//16	13//4	49//16
K2	high	continue	1	false	0//1	4//1	4//1
K2	high	scale	2	true	3//16	23//4	89//16

S4.4 Safe compression and capacity correspondences

The following tables retain every source and every optimizer tie. Masks use the coordinate order (I, A, B, D); semicolon-separated entries are tied solutions, not a sequence of selections. The primary burden keeps $w_I = w_{s_0} = 0$. The committed canonical resource artifacts use the separately displayed total $\widetilde{W}(L) = 1 + W(L)$, so their machine fields `source_burden`, `capacity`, `optimal_burdens`, and `selected_burden` are display burdens. Consequently the capacity objective is $\widetilde{V}^\star(b; \widetilde{B})$ and the penalized net-value field is $\widetilde{J}^\star(b; \lambda) = J^\star(b; \lambda) - \lambda$. Absolute burden reductions and all switching prices are invariant to this common-unit translation. The capacity theorem and Lean formalization use W, not $\widetilde{W}$.

Table 8: Complete safe-compression optimizer correspondence.

Schedule	Source mask	Source library	Display source burden $\widetilde{W}$	Feasible masks	Minimum display burden $\widetilde{W}$	Optimal masks	Optimal libraries	Burden reduction $\Delta\widetilde{W} = \Delta W$	Frontier	Closure	Ties complete
equal_active	0	I	1//1	0	1//1	0	I	0//1	0//1;0//1		true
equal_active	1	I+A	2//1	1	2//1	1	I+A	0//1	0//1;0//1	1	true
equal_active	2	I+B	2//1	2	2//1	2	I+B	0//1	0//1;0//1	1	true
equal_active	3	I+A+B	3//1	1;2;3	2//1	1;2	I+A;I+B	1//1	0//1;0//1	1	true
equal_active	4	I+D	2//1	4	2//1	4	I+D	0//1	2//1;4//1	1	true
equal_active	5	I+A+D	3//1	4;5	2//1	4	I+D	1//1	2//1;4//1	1	true
equal_active	6	I+B+D	3//1	4;6	2//1	4	I+D	1//1	2//1;4//1	1	true
equal_active	7	I+A+B+D	4//1	4;5;6;7	2//1	4	I+D	2//1	2//1;4//1	1	true
carrier_heavy	0	I	1//1	0	1//1	0	I	0//1	0//1;0//1		true
carrier_heavy	1	I+A	3//1	1	3//1	1	I+A	0//1	0//1;0//1	1	true
carrier_heavy	2	I+B	3//1	2	3//1	2	I+B	0//1	0//1;0//1	1	true
carrier_heavy	3	I+A+B	5//1	1;2;3	3//1	1;2	I+A;I+B	2//1	0//1;0//1	1	true
carrier_heavy	4	I+D	2//1	4	2//1	4	I+D	0//1	2//1;4//1	1	true
carrier_heavy	5	I+A+D	4//1	4;5	2//1	4	I+D	2//1	2//1;4//1	1	true
carrier_heavy	6	I+B+D	4//1	4;6	2//1	4	I+D	2//1	2//1;4//1	1	true
carrier_heavy	7	I+A+B+D	6//1	4;5;6;7	2//1	4	I+D	4//1	2//1;4//1	1	true
descendant_heavy	0	I	1//1	0	1//1	0	I	0//1	0//1;0//1		true
descendant_heavy	1	I+A	2//1	1	2//1	1	I+A	0//1	0//1;0//1	1	true
descendant_heavy	2	I+B	2//1	2	2//1	2	I+B	0//1	0//1;0//1	1	true
descendant_heavy	3	I+A+B	3//1	1;2;3	2//1	1;2	I+A;I+B	1//1	0//1;0//1	1	true
descendant_heavy	4	I+D	3//1	4	3//1	4	I+D	0//1	2//1;4//1	1	true
descendant_heavy	5	I+A+D	4//1	4;5	3//1	4	I+D	1//1	2//1;4//1	1	true
descendant_heavy	6	I+B+D	4//1	4;6	3//1	4	I+D	1//1	2//1;4//1	1	true
descendant_heavy	7	I+A+B+D	5//1	4;5;6;7	3//1	4	I+D	2//1	2//1;4//1	1	true

Table 9: Complete capacity paths and value certificates.

Schedule	Belief	Display capacity $\widetilde{B}$	$\widetilde{V}^\star$	Optimal masks	Optimal libraries	Optimal display burdens $\widetilde{W}$	Selected library	Op. value	Gen. value	Total value	Ties complete
equal_active	low	1//1	115//288	0	I	1//1	I	0//1	115//288	115//288	true
equal_active	low	2//1	14//3	4	I+D	2//1	I+D	14//3	0//1	14//3	true
equal_active	low	3//1	14//3	4;5;6	I+D;I+A+D;I+B+D	2//1;3//1;3//1	I+D	14//3	0//1	14//3	true
equal_active	low	4//1	14//3	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;3//1;3//1;4//1	I+D	14//3	0//1	14//3	true
equal_active	high	1//1	23//288	0	I	1//1	I	0//1	23//288	23//288	true
equal_active	high	2//1	22//3	4	I+D	2//1	I+D	22//3	0//1	22//3	true
equal_active	high	3//1	22//3	4;5;6	I+D;I+A+D;I+B+D	2//1;3//1;3//1	I+D	22//3	0//1	22//3	true
equal_active	high	4//1	22//3	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;3//1;3//1;4//1	I+D	22//3	0//1	22//3	true
carrier_heavy	low	1//1	115//288	0	I	1//1	I	0//1	115//288	115//288	true
carrier_heavy	low	2//1	14//3	4	I+D	2//1	I+D	14//3	0//1	14//3	true
carrier_heavy	low	3//1	14//3	4	I+D	2//1	I+D	14//3	0//1	14//3	true
carrier_heavy	low	4//1	14//3	4;5;6	I+D;I+A+D;I+B+D	2//1;4//1;4//1	I+D	14//3	0//1	14//3	true
carrier_heavy	low	5//1	14//3	4;5;6	I+D;I+A+D;I+B+D	2//1;4//1;4//1	I+D	14//3	0//1	14//3	true
carrier_heavy	low	6//1	14//3	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;4//1;4//1;6//1	I+D	14//3	0//1	14//3	true
carrier_heavy	high	1//1	23//288	0	I	1//1	I	0//1	23//288	23//288	true
carrier_heavy	high	2//1	22//3	4	I+D	2//1	I+D	22//3	0//1	22//3	true
carrier_heavy	high	3//1	22//3	4	I+D	2//1	I+D	22//3	0//1	22//3	true
carrier_heavy	high	4//1	22//3	4;5;6	I+D;I+A+D;I+B+D	2//1;4//1;4//1	I+D	22//3	0//1	22//3	true
carrier_heavy	high	5//1	22//3	4;5;6	I+D;I+A+D;I+B+D	2//1;4//1;4//1	I+D	22//3	0//1	22//3	true
carrier_heavy	high	6//1	22//3	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;4//1;4//1;6//1	I+D	22//3	0//1	22//3	true
descendant_heavy	low	1//1	115//288	0	I	1//1	I	0//1	115//288	115//288	true
descendant_heavy	low	2//1	1//1	1;2	I+A;I+B	2//1;2//1	I+A	0//1	1//1	1//1	true
descendant_heavy	low	3//1	14//3	4	I+D	3//1	I+D	14//3	0//1	14//3	true
descendant_heavy	low	4//1	14//3	4;5;6	I+D;I+A+D;I+B+D	3//1;4//1;4//1	I+D	14//3	0//1	14//3	true
descendant_heavy	low	5//1	14//3	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	3//1;4//1;4//1;5//1	I+D	14//3	0//1	14//3	true
descendant_heavy	high	1//1	23//288	0	I	1//1	I	0//1	23//288	23//288	true
descendant_heavy	high	2//1	7//6	1;2	I+A;I+B	2//1;2//1	I+A	0//1	7//6	7//6	true
descendant_heavy	high	3//1	22//3	4	I+D	3//1	I+D	22//3	0//1	22//3	true
descendant_heavy	high	4//1	22//3	4;5;6	I+D;I+A+D;I+B+D	3//1;4//1;4//1	I+D	22//3	0//1	22//3	true
descendant_heavy	high	5//1	22//3	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	3//1;4//1;4//1;5//1	I+D	22//3	0//1	22//3	true

S4.5 Penalized intervals and pairwise crossings

The interval record retains breakpoint ties and the open and terminal cells of each finite affine envelope. The crossing record then lists every pair of branches, including coincident and parallel pairs, nonnegative intersections that never reach the upper envelope, and the globally active breakpoints.

Table 10: Complete penalized-envelope interval and breakpoint record.

Schedule	Belief	Cell	Lower	Upper	Low incl.	Up incl.	Probe	Optimal masks	Optimal libraries	Optimal display burdens $\widetilde{W}$	Selected	$\widetilde{J}^{\star}$	Ties complete
equal_active	low	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;3//1;3//1;4//1	I+D	14//3	true
equal_active	low	open_interval	0//1	1229//288	false	false	1229//576	4	I+D	2//1	I+D	115//288	true
equal_active	low	breakpoint	1229//288	1229//288	true	true	1229//288	0;4	I;I+D	1//1;2//1	I	-557//144	true
equal_active	low	terminal_interval	1229//288		false	false	1517//288	0	I	1//1	I	-701//144	true
equal_active	high	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;3//1;3//1;4//1	I+D	22//3	true
equal_active	high	open_interval	0//1	2089//288	false	false	2089//576	4	I+D	2//1	I+D	23//288	true
equal_active	high	breakpoint	2089//288	2089//288	true	true	2089//288	0;4	I;I+D	1//1;2//1	I	-1033//144	true
equal_active	high	terminal_interval	2089//288		false	false	2377//288	0	I	1//1	I	-1177//144	true
carrier_heavy	low	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;4//1;4//1;6//1	I+D	14//3	true
carrier_heavy	low	open_interval	0//1	1229//288	false	false	1229//576	4	I+D	2//1	I+D	115//288	true
carrier_heavy	low	breakpoint	1229//288	1229//288	true	true	1229//288	0;4	I;I+D	1//1;2//1	I	-557//144	true
carrier_heavy	low	terminal_interval	1229//288		false	false	1517//288	0	I	1//1	I	-701//144	true
carrier_heavy	high	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	2//1;4//1;4//1;6//1	I+D	22//3	true
carrier_heavy	high	open_interval	0//1	2089//288	false	false	2089//576	4	I+D	2//1	I+D	23//288	true
carrier_heavy	high	breakpoint	2089//288	2089//288	true	true	2089//288	0;4	I;I+D	1//1;2//1	I	-1033//144	true
carrier_heavy	high	terminal_interval	2089//288		false	false	2377//288	0	I	1//1	I	-1177//144	true
descendant_heavy	low	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	3//1;4//1;4//1;5//1	I+D	14//3	true
descendant_heavy	low	open_interval	0//1	1229//576	false	false	1229//1152	4	I+D	3//1	I+D	563//384	true
descendant_heavy	low	breakpoint	1229//576	1229//576	true	true	1229//576	0;4	I;I+D	1//1;3//1	I	-111//64	true
descendant_heavy	low	terminal_interval	1229//576		false	false	1805//576	0	I	1//1	I	-175//64	true
descendant_heavy	high	breakpoint	0//1	0//1	true	true	0//1	4;5;6;7	I+D;I+A+D;I+B+D;I+A+B+D	3//1;4//1;4//1;5//1	I+D	22//3	true
descendant_heavy	high	open_interval	0//1	2089//576	false	false	2089//1152	4	I+D	3//1	I+D	727//384	true
descendant_heavy	high	breakpoint	2089//576	2089//576	true	true	2089//576	0;4	I;I+D	1//1;3//1	I	-227//64	true
descendant_heavy	high	terminal_interval	2089//576		false	false	2665//576	0	I	1//1	I	-291//64	true

Table 11: All pairwise resource-price relationships and crossing candidates.

Schedule	Belief	Left library	Left value	Left display burden $\widetilde{W}$	Right library	Right value	Right display burden $\widetilde{W}$	Relation	Price	Nonnegative	Globally active
equal_active	low	I	115//288	1//1	I+A	1//1	2//1	intersection	173//288	true	false
equal_active	low	I	115//288	1//1	I+B	1//1	2//1	intersection	173//288	true	false
equal_active	low	I	115//288	1//1	I+A+B	1//1	3//1	intersection	173//576	true	false
equal_active	low	I	115//288	1//1	I+D	14//3	2//1	intersection	1229//288	true	true
equal_active	low	I	115//288	1//1	I+A+D	14//3	3//1	intersection	1229//576	true	false
equal_active	low	I	115//288	1//1	I+B+D	14//3	3//1	intersection	1229//576	true	false
equal_active	low	I	115//288	1//1	I+A+B+D	14//3	4//1	intersection	1229//864	true	false
equal_active	low	I+A	1//1	2//1	I+B	1//1	2//1	coincident		false	false
equal_active	low	I+A	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
equal_active	low	I+A	1//1	2//1	I+D	14//3	2//1	parallel		false	false
equal_active	low	I+A	1//1	2//1	I+A+D	14//3	3//1	intersection	11//3	true	false
equal_active	low	I+A	1//1	2//1	I+B+D	14//3	3//1	intersection	11//3	true	false
equal_active	low	I+A	1//1	2//1	I+A+B+D	14//3	4//1	intersection	11//6	true	false
equal_active	low	I+B	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
equal_active	low	I+B	1//1	2//1	I+D	14//3	2//1	parallel		false	false
equal_active	low	I+B	1//1	2//1	I+A+D	14//3	3//1	intersection	11//3	true	false
equal_active	low	I+B	1//1	2//1	I+B+D	14//3	3//1	intersection	11//3	true	false
equal_active	low	I+B	1//1	2//1	I+A+B+D	14//3	4//1	intersection	11//6	true	false
equal_active	low	I+A+B	1//1	3//1	I+D	14//3	2//1	intersection	-11//3	false	false
equal_active	low	I+A+B	1//1	3//1	I+A+D	14//3	3//1	parallel		false	false
equal_active	low	I+A+B	1//1	3//1	I+B+D	14//3	3//1	parallel		false	false
equal_active	low	I+A+B	1//1	3//1	I+A+B+D	14//3	4//1	intersection	11//3	true	false
equal_active	low	I+D	14//3	2//1	I+A+D	14//3	3//1	intersection	0//1	true	true
equal_active	low	I+D	14//3	2//1	I+B+D	14//3	3//1	intersection	0//1	true	true
equal_active	low	I+D	14//3	2//1	I+A+B+D	14//3	4//1	intersection	0//1	true	true
equal_active	low	I+A+D	14//3	3//1	I+B+D	14//3	3//1	coincident		false	false
equal_active	low	I+A+D	14//3	3//1	I+A+B+D	14//3	4//1	intersection	0//1	true	true
equal_active	low	I+B+D	14//3	3//1	I+A+B+D	14//3	4//1	intersection	0//1	true	true
equal_active	high	I	23//288	1//1	I+A	7//6	2//1	intersection	313//288	true	false
equal_active	high	I	23//288	1//1	I+B	7//6	2//1	intersection	313//288	true	false
equal_active	high	I	23//288	1//1	I+A+B	7//6	3//1	intersection	313//576	true	false
equal_active	high	I	23//288	1//1	I+D	22//3	2//1	intersection	2089//288	true	true
equal_active	high	I	23//288	1//1	I+A+D	22//3	3//1	intersection	2089//576	true	false
equal_active	high	I	23//288	1//1	I+B+D	22//3	3//1	intersection	2089//576	true	false
equal_active	high	I	23//288	1//1	I+A+B+D	22//3	4//1	intersection	2089//864	true	false
equal_active	high	I+A	7//6	2//1	I+B	7//6	2//1	coincident		false	false
equal_active	high	I+A	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false
equal_active	high	I+A	7//6	2//1	I+D	22//3	2//1	parallel		false	false
equal_active	high	I+A	7//6	2//1	I+A+D	22//3	3//1	intersection	37//6	true	false
equal_active	high	I+A	7//6	2//1	I+B+D	22//3	3//1	intersection	37//6	true	false
equal_active	high	I+A	7//6	2//1	I+A+B+D	22//3	4//1	intersection	37//12	true	false
equal_active	high	I+B	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false
equal_active	high	I+B	7//6	2//1	I+D	22//3	2//1	parallel		false	false
equal_active	high	I+B	7//6	2//1	I+A+D	22//3	3//1	intersection	37//6	true	false

Schedule	Belief	Left library	Left value	Left display burden $\widetilde{W}$	Right library	Right value	Right display burden $\widetilde{W}$	Relation	Price	Nonnegative	Globally active
equal_active	high	I+B	7//6	2//1	I+B+D	22//3	3//1	intersection	37//6	true	false
equal_active	high	I+B	7//6	2//1	I+A+B+D	22//3	4//1	intersection	37//12	true	false
equal_active	high	I+A+B	7//6	3//1	I+D	22//3	2//1	intersection	-37//6	false	false
equal_active	high	I+A+B	7//6	3//1	I+A+D	22//3	3//1	parallel		false	false
equal_active	high	I+A+B	7//6	3//1	I+B+D	22//3	3//1	parallel		false	false
equal_active	high	I+A+B	7//6	3//1	I+A+B+D	22//3	4//1	intersection	37//6	true	false
equal_active	high	I+D	22//3	2//1	I+A+D	22//3	3//1	intersection	0//1	true	true
equal_active	high	I+D	22//3	2//1	I+B+D	22//3	3//1	intersection	0//1	true	true
equal_active	high	I+D	22//3	2//1	I+A+B+D	22//3	4//1	intersection	0//1	true	true
equal_active	high	I+A+D	22//3	3//1	I+B+D	22//3	3//1	coincident		false	false
equal_active	high	I+A+D	22//3	3//1	I+A+B+D	22//3	4//1	intersection	0//1	true	true
equal_active	high	I+B+D	22//3	3//1	I+A+B+D	22//3	4//1	intersection	0//1	true	true
carrier_heavy	low	I	115//288	1//1	I+A	1//1	3//1	intersection	173//576	true	false
carrier_heavy	low	I	115//288	1//1	I+B	1//1	3//1	intersection	173//576	true	false
carrier_heavy	low	I	115//288	1//1	I+A+B	1//1	5//1	intersection	173//1152	true	false
carrier_heavy	low	I	115//288	1//1	I+D	14//3	2//1	intersection	1229//288	true	true
carrier_heavy	low	I	115//288	1//1	I+A+D	14//3	4//1	intersection	1229//864	true	false
carrier_heavy	low	I	115//288	1//1	I+B+D	14//3	4//1	intersection	1229//864	true	false
carrier_heavy	low	I	115//288	1//1	I+A+B+D	14//3	6//1	intersection	1229//1440	true	false
carrier_heavy	low	I+A	1//1	3//1	I+B	1//1	3//1	coincident		false	false
carrier_heavy	low	I+A	1//1	3//1	I+A+B	1//1	5//1	intersection	0//1	true	false
carrier_heavy	low	I+A	1//1	3//1	I+D	14//3	2//1	intersection	-11//3	false	false
carrier_heavy	low	I+A	1//1	3//1	I+A+D	14//3	4//1	intersection	11//3	true	false
carrier_heavy	low	I+A	1//1	3//1	I+B+D	14//3	4//1	intersection	11//3	true	false
carrier_heavy	low	I+A	1//1	3//1	I+A+B+D	14//3	6//1	intersection	11//9	true	false
carrier_heavy	low	I+B	1//1	3//1	I+A+B	1//1	5//1	intersection	0//1	true	false
carrier_heavy	low	I+B	1//1	3//1	I+D	14//3	2//1	intersection	-11//3	false	false
carrier_heavy	low	I+B	1//1	3//1	I+A+D	14//3	4//1	intersection	11//3	true	false
carrier_heavy	low	I+B	1//1	3//1	I+B+D	14//3	4//1	intersection	11//3	true	false
carrier_heavy	low	I+B	1//1	3//1	I+A+B+D	14//3	6//1	intersection	11//9	true	false
carrier_heavy	low	I+A+B	1//1	5//1	I+D	14//3	2//1	intersection	-11//9	false	false
carrier_heavy	low	I+A+B	1//1	5//1	I+A+D	14//3	4//1	intersection	-11//3	false	false
carrier_heavy	low	I+A+B	1//1	5//1	I+B+D	14//3	4//1	intersection	-11//3	false	false
carrier_heavy	low	I+A+B	1//1	5//1	I+A+B+D	14//3	6//1	intersection	11//3	true	false
carrier_heavy	low	I+D	14//3	2//1	I+A+D	14//3	4//1	intersection	0//1	true	true
carrier_heavy	low	I+D	14//3	2//1	I+B+D	14//3	4//1	intersection	0//1	true	true
carrier_heavy	low	I+D	14//3	2//1	I+A+B+D	14//3	6//1	intersection	0//1	true	true
carrier_heavy	low	I+A+D	14//3	4//1	I+B+D	14//3	4//1	coincident		false	false
carrier_heavy	low	I+A+D	14//3	4//1	I+A+B+D	14//3	6//1	intersection	0//1	true	true
carrier_heavy	low	I+B+D	14//3	4//1	I+A+B+D	14//3	6//1	intersection	0//1	true	true
carrier_heavy	high	I	23//288	1//1	I+A	7//6	3//1	intersection	313//576	true	false
carrier_heavy	high	I	23//288	1//1	I+B	7//6	3//1	intersection	313//576	true	false
carrier_heavy	high	I	23//288	1//1	I+A+B	7//6	5//1	intersection	313//1152	true	false
carrier_heavy	high	I	23//288	1//1	I+D	22//3	2//1	intersection	2089//288	true	true
carrier_heavy	high	I	23//288	1//1	I+A+D	22//3	4//1	intersection	2089//864	true	false
carrier_heavy	high	I	23//288	1//1	I+B+D	22//3	4//1	intersection	2089//864	true	false

Schedule	Belief	Left library	Left value	Left display burden $\widetilde{W}$	Right library	Right value	Right display burden $\widetilde{W}$	Relation	Price	Nonnegative	Globally active
carrier_heavy	high	I	23//288	1//1	I+A+B+D	22//3	6//1	intersection	2089//1440	true	false
carrier_heavy	high	I+A	7//6	3//1	I+B	7//6	3//1	coincident		false	false
carrier_heavy	high	I+A	7//6	3//1	I+A+B	7//6	5//1	intersection	0//1	true	false
carrier_heavy	high	I+A	7//6	3//1	I+D	22//3	2//1	intersection	-37//6	false	false
carrier_heavy	high	I+A	7//6	3//1	I+A+D	22//3	4//1	intersection	37//6	true	false
carrier_heavy	high	I+A	7//6	3//1	I+B+D	22//3	4//1	intersection	37//6	true	false
carrier_heavy	high	I+A	7//6	3//1	I+A+B+D	22//3	6//1	intersection	37//18	true	false
carrier_heavy	high	I+B	7//6	3//1	I+A+B	7//6	5//1	intersection	0//1	true	false
carrier_heavy	high	I+B	7//6	3//1	I+D	22//3	2//1	intersection	-37//6	false	false
carrier_heavy	high	I+B	7//6	3//1	I+A+D	22//3	4//1	intersection	37//6	true	false
carrier_heavy	high	I+B	7//6	3//1	I+B+D	22//3	4//1	intersection	37//6	true	false
carrier_heavy	high	I+B	7//6	3//1	I+A+B+D	22//3	6//1	intersection	37//18	true	false
carrier_heavy	high	I+A+B	7//6	5//1	I+D	22//3	2//1	intersection	-37//18	false	false
carrier_heavy	high	I+A+B	7//6	5//1	I+A+D	22//3	4//1	intersection	-37//6	false	false
carrier_heavy	high	I+A+B	7//6	5//1	I+B+D	22//3	4//1	intersection	-37//6	false	false
carrier_heavy	high	I+A+B	7//6	5//1	I+A+B+D	22//3	6//1	intersection	37//6	true	false
carrier_heavy	high	I+D	22//3	2//1	I+A+D	22//3	4//1	intersection	0//1	true	true
carrier_heavy	high	I+D	22//3	2//1	I+B+D	22//3	4//1	intersection	0//1	true	true
carrier_heavy	high	I+D	22//3	2//1	I+A+B+D	22//3	6//1	intersection	0//1	true	true
carrier_heavy	high	I+A+D	22//3	4//1	I+B+D	22//3	4//1	coincident		false	false
carrier_heavy	high	I+A+D	22//3	4//1	I+A+B+D	22//3	6//1	intersection	0//1	true	true
carrier_heavy	high	I+B+D	22//3	4//1	I+A+B+D	22//3	6//1	intersection	0//1	true	true
descendant_heavy	low	I	115//288	1//1	I+A	1//1	2//1	intersection	173//288	true	false
descendant_heavy	low	I	115//288	1//1	I+B	1//1	2//1	intersection	173//288	true	false
descendant_heavy	low	I	115//288	1//1	I+A+B	1//1	3//1	intersection	173//576	true	false
descendant_heavy	low	I	115//288	1//1	I+D	14//3	3//1	intersection	1229//576	true	true
descendant_heavy	low	I	115//288	1//1	I+A+D	14//3	4//1	intersection	1229//864	true	false
descendant_heavy	low	I	115//288	1//1	I+B+D	14//3	4//1	intersection	1229//864	true	false
descendant_heavy	low	I	115//288	1//1	I+A+B+D	14//3	5//1	intersection	1229//1152	true	false
descendant_heavy	low	I+A	1//1	2//1	I+B	1//1	2//1	coincident		false	false
descendant_heavy	low	I+A	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
descendant_heavy	low	I+A	1//1	2//1	I+D	14//3	3//1	intersection	11//3	true	false
descendant_heavy	low	I+A	1//1	2//1	I+A+D	14//3	4//1	intersection	11//6	true	false
descendant_heavy	low	I+A	1//1	2//1	I+B+D	14//3	4//1	intersection	11//6	true	false
descendant_heavy	low	I+A	1//1	2//1	I+A+B+D	14//3	5//1	intersection	11//9	true	false
descendant_heavy	low	I+B	1//1	2//1	I+A+B	1//1	3//1	intersection	0//1	true	false
descendant_heavy	low	I+B	1//1	2//1	I+D	14//3	3//1	intersection	11//3	true	false
descendant_heavy	low	I+B	1//1	2//1	I+A+D	14//3	4//1	intersection	11//6	true	false
descendant_heavy	low	I+B	1//1	2//1	I+B+D	14//3	4//1	intersection	11//6	true	false
descendant_heavy	low	I+B	1//1	2//1	I+A+B+D	14//3	5//1	intersection	11//9	true	false
descendant_heavy	low	I+A+B	1//1	3//1	I+D	14//3	3//1	parallel		false	false
descendant_heavy	low	I+A+B	1//1	3//1	I+A+D	14//3	4//1	intersection	11//3	true	false
descendant_heavy	low	I+A+B	1//1	3//1	I+B+D	14//3	4//1	intersection	11//3	true	false
descendant_heavy	low	I+A+B	1//1	3//1	I+A+B+D	14//3	5//1	intersection	11//6	true	false
descendant_heavy	low	I+D	14//3	3//1	I+A+D	14//3	4//1	intersection	0//1	true	true
descendant_heavy	low	I+D	14//3	3//1	I+B+D	14//3	4//1	intersection	0//1	true	true

Schedule	Belief	Left library	Left value	Left display burden $\widetilde{W}$	Right library	Right value	Right display burden $\widetilde{W}$	Relation	Price	Nonnegative	Globally active
descendant_heavy	low	I+D	14//3	3//1	I+A+B+D	14//3	5//1	intersection	0//1	true	true
descendant_heavy	low	I+A+D	14//3	4//1	I+B+D	14//3	4//1	coincident		false	false
descendant_heavy	low	I+A+D	14//3	4//1	I+A+B+D	14//3	5//1	intersection	0//1	true	true
descendant_heavy	low	I+B+D	14//3	4//1	I+A+B+D	14//3	5//1	intersection	0//1	true	true
descendant_heavy	high	I	23//288	1//1	I+A	7//6	2//1	intersection	313//288	true	false
descendant_heavy	high	I	23//288	1//1	I+B	7//6	2//1	intersection	313//288	true	false
descendant_heavy	high	I	23//288	1//1	I+A+B	7//6	3//1	intersection	313//576	true	false
descendant_heavy	high	I	23//288	1//1	I+D	22//3	3//1	intersection	2089//576	true	true
descendant_heavy	high	I	23//288	1//1	I+A+D	22//3	4//1	intersection	2089//864	true	false
descendant_heavy	high	I	23//288	1//1	I+B+D	22//3	4//1	intersection	2089//864	true	false
descendant_heavy	high	I	23//288	1//1	I+A+B+D	22//3	5//1	intersection	2089//1152	true	false
descendant_heavy	high	I+A	7//6	2//1	I+B	7//6	2//1	coincident		false	false
descendant_heavy	high	I+A	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false
descendant_heavy	high	I+A	7//6	2//1	I+D	22//3	3//1	intersection	37//6	true	false
descendant_heavy	high	I+A	7//6	2//1	I+A+D	22//3	4//1	intersection	37//12	true	false
descendant_heavy	high	I+A	7//6	2//1	I+B+D	22//3	4//1	intersection	37//12	true	false
descendant_heavy	high	I+A	7//6	2//1	I+A+B+D	22//3	5//1	intersection	37//18	true	false
descendant_heavy	high	I+B	7//6	2//1	I+A+B	7//6	3//1	intersection	0//1	true	false
descendant_heavy	high	I+B	7//6	2//1	I+D	22//3	3//1	intersection	37//6	true	false
descendant_heavy	high	I+B	7//6	2//1	I+A+D	22//3	4//1	intersection	37//12	true	false
descendant_heavy	high	I+B	7//6	2//1	I+B+D	22//3	4//1	intersection	37//12	true	false
descendant_heavy	high	I+B	7//6	2//1	I+A+B+D	22//3	5//1	intersection	37//18	true	false
descendant_heavy	high	I+A+B	7//6	3//1	I+D	22//3	3//1	parallel		false	false
descendant_heavy	high	I+A+B	7//6	3//1	I+A+D	22//3	4//1	intersection	37//6	true	false
descendant_heavy	high	I+A+B	7//6	3//1	I+B+D	22//3	4//1	intersection	37//6	true	false
descendant_heavy	high	I+A+B	7//6	3//1	I+A+B+D	22//3	5//1	intersection	37//12	true	false
descendant_heavy	high	I+D	22//3	3//1	I+A+D	22//3	4//1	intersection	0//1	true	true
descendant_heavy	high	I+D	22//3	3//1	I+B+D	22//3	4//1	intersection	0//1	true	true
descendant_heavy	high	I+D	22//3	3//1	I+A+B+D	22//3	5//1	intersection	0//1	true	true
descendant_heavy	high	I+A+D	22//3	4//1	I+B+D	22//3	4//1	coincident		false	false
descendant_heavy	high	I+A+D	22//3	4//1	I+A+B+D	22//3	5//1	intersection	0//1	true	true
descendant_heavy	high	I+B+D	22//3	4//1	I+A+B+D	22//3	5//1	intersection	0//1	true	true

The committed CSV files retain additional machine-oriented counts and certificate fields not repeated as display columns above. In particular, they record feasible and enumerated counts, all-optima-feasible flags, productive-value preservation, capacity shadow increments, and optimizer fragility diagnostics. These fields remain part of the reproducibility record; the tables here preserve every numerical row and every optimizer or crossing candidate needed to reconstruct the reported correspondences.

S5 Complete Registered Randomized Study

This section preserves the detailed estimands, raw-realizability gates, interaction diagnostics, sequential-prefix records, and frozen-pilot comparison behind the randomized results summarized in the main paper. The study is a synthetic structural stress test of a fixed finite generator, not a population sample or a population-frequency estimate.

S5.1 Synthetic estimands and exact fixture bridge

For fixture f, horizon H, and declared input set X_f, define

$$\epsilon^{\text{proj}}_{f,H} := \max_{(b,L)\in X_f} |V^{\text{raw}}_H(b, L) - V^K_H(b, \phi(L))|.$$

For deletion $L \to L^-$, frontier-only pruning $L \to L^F$, and insertion, the remaining exact-fixture estimands are

$$\epsilon_F := \|F_L - F_{L^-}\|_\infty, \quad \epsilon_C := \mathbf{1}\{C_L \neq C_{L^-}\}, \quad \epsilon_V := \max_b |V_H(b, L) - V_H(b, L^-)|,$$

$$\text{Loss}^F_H := V_H(b, L) - V_H(b, L^F), \quad \lambda^F := \frac{\text{Loss}^F_H}{\beta^d \rho^d \pi C - \kappa}, \quad \epsilon^{\text{dec}} := \Delta^{\text{tot}} - \Delta^{\text{op}} - \Delta^{\text{gen}}.$$

The ratio λ^F is used only when its denominator is positive. For a four-corner value fixture and a one-shot cost-covering set, define

$$J := [V(F_1, C_1) - V(F_1, C_0)] - [V(F_0, C_1) - V(F_0, C_0)], \quad \mathcal{R} := \{b : \Psi(b) - \kappa(b) \geq 0\},$$

and let $c(\mathcal{R})$ be the number of connected components on the ordered belief grid.

The exact fixture outcomes are retained here as a bridge between those estimands and the randomized design.

Mechanism	Registered exact input	Computed diagnostic
Safe compression	Four-strategy, two-belief library	Minimum and greedy sizes both 3; frontier, closure, and value loss all zero
Frontier-only loss	Future rewards $0, 2, 4, 10, 20, 40$	Loss $R/2$ in all six cases; largest displayed loss 20
Value decomposition	Operational-only, generative-only, and joint insertions	$(\Delta^{\text{op}}, \Delta^{\text{gen}}) = (3, 0), (0, 4), (3, 1)$
Monotone single gap	Potential $(0, 1/2, 3/2, 3)$	Upper-grid research set; one component
Single peak, bad kernel	Gap $(0, 1, 0)$, potential $(1, 0, 1)$	Preservation assumption fails; two components
Separated multi-gap	Potential $(4, 41/32, 1/2, 41/32, 4)$	Constant-cost research set has two components

Table 12: Selected-instance outcomes from the exact mechanism fixtures.

The scaled frontier-only family obtains loss $R/2$ at each $R \in \{0, 2, 4, 10, 20, 40\}$. The unit-cap fixture is the normalized result; the bad-kernel and separated-gap fixtures retain their disconnection diagnostics.

S5.2 Registered design and randomized estimands

The randomized layer uses master seed `6075990691714899803` and all $N = 1024$ registered horizon-four trials. The complete 2^7 factorial crosses sparse/dense frontiers, low/high module overlap, weak/strong module complementarity, low/high project cost, short/long duration, low/high admission, and low/high persistence. Each of the 128 cells has eight replicates. The complete registry records 1,024 trial, catalog, project, and deletion seeds; all 4,096 are distinct. The initial design, sequential-precision amendment, and executable-protocol amendment were locked before outcomes.

For trial t, source library L_t, pruning method m, and pruned library L_{tm}, let

$$W_t(L) := \text{passive value}, \qquad \Omega_t(L) := V_t(L) - W_t(L),$$

$$\begin{aligned} L_{tm}^{\mathrm{op}} &:= W_t(L_t) - W_t(L_{tm}), & L_{tm}^{\mathrm{gen}} &:= \Omega_t(L_t) - \Omega_t(L_{tm}), \\ L_{tm}^{\mathrm{tot}} &:= V_t(L_t) - V_t(L_{tm}) = L_{tm}^{\mathrm{op}} + L_{tm}^{\mathrm{gen}}, & r_{tm} &:= 1 - |L_{tm}|/|L_t|. \end{aligned}$$

For frontier-only pruning F, define

$$\begin{aligned} \widehat{p}_F &:= N^{-1} \sum_t \mathbf{1}\{L_{tF}^{\mathrm{tot}} > 0\}, \quad \overline{L}_F^{+|+} := \frac{\sum_t [L_{tF}^{\mathrm{tot}}]_+}{\sum_t \mathbf{1}\{L_{tF}^{\mathrm{tot}} > 0\}}, \\ \widetilde{L}_F^{\max} &:= \max_t \frac{[L_{tF}^{\mathrm{tot}}]_+}{V_t(L_t)}. \end{aligned}$$

Strict positivity of every source value is a construction gate. An operationally silent generative asset $s \in L_t$ satisfies

$$F_{L_t} = F_{L_t \setminus \{s\}}, \qquad C_{L_t} \neq C_{L_t \setminus \{s\}}, \qquad V_t(L_t) - V_t(L_t \setminus \{s\}) > 0.$$

Its asset-level frequency uses every noninactive source asset; a separate library-level count records whether any such asset is present.

Every interaction observation starts from four actual raw libraries:

$$\begin{aligned} L_{00} &= L_{\mathrm{base}}, & L_{10} &= L_{\mathrm{base}} \cup A_F, \\ L_{01} &= L_{\mathrm{base}} \cup A_C, & L_{11} &= L_{\mathrm{base}} \cup A_F \cup A_C, \end{aligned}$$

where A_F is closure-silent and frontier-improving, A_C is frontier-silent and closure-improving, and the two additions commute. All four libraries use the same catalog and module closure. Their frontiers, closures, project laws, menus, transitions, and values are recomputed from raw primitives. With J_t defined by the four-corner difference above, the sign frequencies are

$$\widehat{p}_{\mathrm{sub}} := \frac{\#\{t : J_t < 0\}}{1024}, \qquad \widehat{p}_{\mathrm{comp}} := \frac{\#\{t : J_t > 0\}}{1024},$$

with the analogous $\widehat{p}_0$ for $J_t = 0$. No synthetic compressed rectangle enters these denominators.

All profiles, module rows, closure tables, kernels, project requirements, admitted laws, raw updates, passive values, and Bellman values use `Rational{BigInt}`. Before writing any artifact, hard gates require all 1,024 trials, all 4,096 raw corners, and every raw library to be valid; every compressed state to have an enumerated raw witness; all raw/compressed values to agree at every belief and remaining horizon $0, \ldots, 4$; frontier-only passive loss to be zero; innovation-safe frontier, closure, operational, generative, and total loss to be exactly zero; every signed decomposition to close; and every displayed theorem-condition flag to be computed from exact laws and action tables. The completed audit records 47,458 raw-witness rows and 1,920 raw/compressed value comparisons per trial.

Registered estimand	Exact count	Exact estimate
Positive frontier-only dynamic loss	398/1024	199/512
Conditional mean positive loss	398 positive trials	124909129/78249984
Maximum normalized loss	1024 trials	84963/222163
Positive innovation-safe loss	0/1024	0
Mean frontier-only compression ratio	1024 trials	5/12
Mean innovation-safe compression ratio	1024 trials	1/8
Silent generative asset	388/5120	97/1280
Substitution, $J_t < 0$	256/1024	1/4
Complementarity, $J_t > 0$	141/1024	141/1024
Zero interaction, $J_t = 0$	627/1024	627/1024

Table 13: Registered finite-generator outcomes. Ratios are structural stress-test frequencies, not estimates of real-population prevalence.

S5.3 Registered outcomes and factor slices

Frontier-only pruning has positive loss in 398 trials; conditional on positive loss, its mean magnitude is 124909129/78249984 $\approx$ 1.5963, and the largest normalized loss is 84963/222163 $\approx$ 0.3824. Innovation-safe pruning has exact zero frontier, closure, operational, generative, and total loss in every trial. Its smaller mean compression ratio reflects the additional closure- preservation constraint. Silent generative carriers occur in 388 of 5,120 noninactive source-asset observations.

The primitive common-gap predicate is the mechanical conjunction of frontier-independent project primitives and menus, poor-to-rich menu inclusion, an antitone recursive descendant gap, nonnegative rich exposure, zero poor exposure, and exact action-value identities. It holds in 512 trials. Those rows contain 255 substitutions, 257 zeros, and no complementarity. The other 512 rows are registered boundary constructions: the 256 frontier-dependent-generator rows contain all 141 complementarities, one substitution, and 114 zeros; the 256 positive-poor-exposure rows are all zero. Hence every positive J_t occurs where frontier-independent generation and completion fail. Marginally, complementarity is more frequent under low cost than high cost (97 versus 44), short than long duration (107 versus 34), and high than low admission (95 versus 46). These balanced factor slices locate generated cases; they are neither causal comparisons nor a theorem that complementarity is rare.

The companion exact interaction response preserves the registered 2,430-row frontier–project boundary grid and five canonical four-corner fixtures: primitive strict substitution, the saturation boundary, optimizer-switching complementarity, frontier-dependent-success complementarity, and nonconstant separability. It crosses strict low/high frontier pairs, closure richness, project cost, admission probability, descendant payoff, incumbent operating reward, project duration, and frontier dependence of generator quality. Each row records the two closure increments, their cross difference, four selected projects, four realizability flags, and the primitive sufficient-condition certificate. Nonrealizable rows remain diagnostic records but are never aggregated into interaction-sign frequencies; Supplement S2 reports the corresponding interaction-theory interpretation.

S5.4 Sequential prefixes and descriptive uncertainty

The sequential amendment fixed cumulative views at $N = 50, 100, 200, 300, 500, 750, 1000, 1024$ before outcomes and required the final estimate to use all 1,024 trials. The early prefixes are not claimed to be factor-balanced.

At $N = 1024$, the descriptive Wilson intervals are $[0.3593, 0.4189]$ for frontier loss, $[0, 0.00374]$

N	Frontier loss	Safe loss	Silent asset	Substitution	Complementarity
50	25/50	0/50	25/250	27/50	0/50
100	45/100	0/100	45/500	47/100	0/100
200	101/200	0/200	101/1000	64/200	39/200
300	153/300	0/300	153/1500	87/300	69/300
500	198/500	0/500	192/2500	127/500	69/500
750	321/750	0/750	315/3750	192/750	130/750
1000	396/1000	0/1000	388/5000	256/1000	141/1000
1024	398/1024	0/1024	388/5120	256/1024	141/1024

Table 14: Cumulative registered event counts. The silent-asset denominator is five noninactive source assets per trial. Prefixes diagnose simulation precision and never select the final sample.

for innovation-safe loss, $[0.06885, 0.08335]$ for silent assets, $[0.2244, 0.2774]$ for substitution, and $[0.1179, 0.1602]$ for complementarity. The corresponding Monte Carlo standard errors are 0.08416 for conditional mean positive loss, 0.00261 for mean frontier compression, and 0.00432 for mean safe compression. Innovation-safe loss has zero events at every prefix and therefore retains the registered sparse-event warning; complementarity has zero events through $N = 100$, illustrating why the registered maximum cannot be selected from an early sign.

N	Positive support	Mean positive loss	Frontier compression	Safe compression
50	25	1818477/1638400	31/75	9/100
100	45	1623911/1474560	5/12	3/25
200	101	5624367/3309568	5/12	51/400
300	153	55259345/30081024	187/450	227/1800
500	198	63735497/38928384	1249/3000	47/375
750	321	107121079/63111168	1873/4500	557/4500
1000	396	124655857/77856768	5/12	47/375
1024	398	124909129/78249984	5/12	1/8

Table 15: Cumulative mean diagnostics, all computed exactly.

The complete 64-row cumulative table, 896-row factor-stratified table, exact sample variances, interval values, warnings, and stabilization figure remain in the registered artifacts indexed in Supplement S7.

S5.5 Frozen pilot and policy-map diagnostics

The earlier $N = 90$ study remains a frozen pilot and is not pooled with the registered v2 estimates. The two generators differ materially, so the comparison below records provenance rather than a trend.

The separately locked resource extension reuses only the frozen v2 trial registry and does not pool its outcomes with the parent study. Every trial enumerates all 32 inactive-containing sublibraries. Complete optimizer correspondences, capacity staircases, active pairwise breakpoints, price paths, and channel arcs are indexed in Supplement S7.

Quantity	Frozen pilot, $N = 90$	Registered v2, $N = 1024$
Design	twelve marginally balanced three-level factors	complete 2^7 factorial, eight replicates per cell
Positive frontier-only loss	8/90	398/1024
Conditional mean positive loss	3705/16384	124909129/78249984
Mean frontier compression	22/45	5/12
Mean innovation-safe compression	2009/5400	1/8
Positive innovation-safe loss	0/90	0/1024
Silent generative assets	3/360	388/5120
Interaction comparison	omitted: no four-raw-witness gate	256/141/627 negative/positive/zero

Table 16: Frozen-pilot and v2 results are reported separately. The pilot interaction statistic is not compared because its synthetic compressed rectangle did not require four raw witnesses.

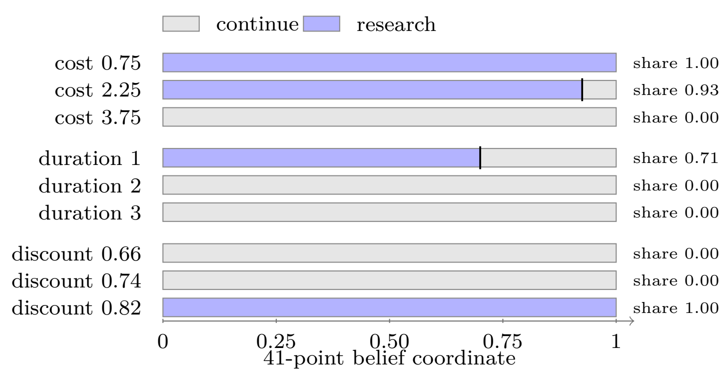

Figure 2: Policy-map slices from the unified positive-duration response experiment. The estimand in each row is the share of 41 enumerated belief points selecting research in a converged raw-derived Float64 solve; gray denotes Continue and blue denotes research. Filled spans summarize the connected finite-grid action set and the boundary line is a grid cutoff, not a continuous root or smooth interpolation. All duration rows are strictly positive, use P_B^d, and apply the active-operation flag.

S6 Full Financial Audits and Secondary Tests

The empirical layer audits ex post whether a financial library can be compressed without discarding access to useful descendants. The primary comparison is innovation-safe versus frontier-only pruning; coverage ranking is a secondary stress test. The information-set boundary is summarized in the main paper's Appendix E and applies to every definition and result below. The terminal financial audit is the fixed-split protocol, whereas the annual walk-forward financial audit is the five-decision protocol. They are separate audits with different held-out estimands; neither supersedes the other and their outcome levels are never pooled.

S6.1 Data, design locks, and resampling

Both audits use the same licensed CRSP/WRDS snapshot; raw and row-level files remain nonredistributable. The fixed daily snapshot covers 2008–2024. The terminal audit design locks 25 date-valid surviving ETFs, 2,400 strategies, three sample splits, 500 seeded circular-block replications with 20-session blocks, and the 1/5/10-basis-point cost schedule before scoring. The annual walk-forward design fixes an outcome-blind 100-ETF manifest, five belief states, trailing five-year estimation, sequential top-five selection, five annual targets, 2,000 seeded annual-unit bootstrap replications, and the same cost schedule. Signals use information through close t, followed by a two-return-index lag. Neither universe is survivorship-free and neither holdout is prospective.

For each ETF, the fixed grammar enumerates three directional signals and two choices each of entry filter, horizon, sizing, exit, and risk constraint: $3 \cdot 2^5 = 96$ policies per instrument, each carrying the resulting seven modules. Development is 2009–2014, validation is 2015–2019, and the held-out period is 2020–2024. The ignored licensed panels contain 106,975 and 427,900 rows, respectively, each with 4,279 complete common dates.

S6.2 Compression and descendant-quality estimands

Write $a \in \{\mathrm{T}, \mathrm{A}\}$ for the locked terminal and annual walk-forward audits, L_a for the source library, L_a^F for frontier-only pruning, and L_a^S for innovation-safe compression. For empirical belief cell b, policy payoff $\widehat{j}_{ai}(b)$, and fixed cell weights $\widehat{\mu}_a(b)$, define

$$\widehat{F}_a(b; L') := \max_{i \in L'} \widehat{j}_{ai}(b), \qquad \overline{F}_a(L') := \sum_b \widehat{\mu}_a(b) \widehat{F}_a(b; L'), \tag{S6.1}$$

$$C_a(L') := \bigcup_{i \in L'} M(i), \qquad r_a(L') := 1 - \frac{|L'|}{|L_a|}. \tag{S6.2}$$

The full frontier vector is computed from validation profiles and is available to both pruning rules; frontier equality is their common deletion gate. Module closure is structural information available at the same time and is the additional innovation-safe gate. The compression ratio is calculated only after a pruning run terminates and is a retrospective, outcome-blind description, not a deletion criterion.

Let $\mathcal{G}_a$ be the fixed candidate catalog, $M(c)$ the modules required by candidate c, and $q_a(c)$ its quality measured from held-out audit outcomes. Define

$$\begin{aligned} E_a(L') &:= \{c \in \mathcal{G}_a : M(c) \subseteq C_a(L')\}, \\ Q_a(L') &:= \max_{c \in E_a(L')} q_a(c). \end{aligned} \tag{S6.3}$$

The enabled set $E_a(L')$ is structurally derivable from the fixed catalog and closure before the holdout, but neither pruning rule tests candidate-set equality. The quantity $Q_a(L')$, the *ex post* *enabled-descendant opportunity quality*, measures whether the modules retained in L' would have preserved access to candidates that realized high quality in the held-out audit. It is evaluated only after pruning decisions are fixed and is neither a forecast, a policy score, nor a deployable selection criterion.

For the locked terminal audit, q_{T} is locked-period candidate net utility. For the annual walk-forward audit, q_{A} is mean realized candidate coverage over the five 2020–2024 targets. Their levels are not comparable. The displayed changes are

$$\Delta\overline{F}_a(L') := \overline{F}_a(L') - \overline{F}_a(L_a), \qquad \Delta Q_a(L') := Q_a(L') - Q_a(L_a).$$

At each deletion decision, innovation-safe pruning accepts a removal using only equality of the current frontier and module closure; it never evaluates Q_a. Frontier-only pruning uses frontier equality alone. Thus ΔQ_a is an ex post mechanism diagnostic, not an allegation that frontier-only pruning ignored known future outcomes.

S6.3 Deletion-contribution audit

For policy i retained in the safe library L_a^S, define the empirical deletion contributions

$$\begin{aligned} \widehat{O}_{ai} &:= \widehat{P}_a(L_a^S) - \widehat{P}_a(L_a^S \setminus \{i\}), \\ \widehat{G}_{ai} &:= [\widehat{U}_a(L_a^S) - \widehat{U}_a(L_a^S \setminus \{i\})] - \widehat{O}_{ai}, \end{aligned} \tag{S6.4}$$

where $\widehat{P}_a$ freezes the library and $\widehat{U}_a$ retains the research option. The operational-only, generative-only, both, and neither counts classify the signs of $(\widehat{O}_{ai}, \widehat{G}_{ai})$. Frontier sparsity, module uniqueness, and candidate dependence are

$$S_{ai} := \mathbf{1}\{\widehat{O}_{ai} = 0\}, \qquad H_{ai} := \frac{1}{|M(i)|} \sum_{m \in M(i)} \mathbf{1}\Big\{|\{j \in L_a^S : m \in M(j)\}| = 1\Big\},$$

$$D_{ai} := |\{c \in \mathcal{G}_a : M(c) \subseteq C_a(L_a^S), M(c) \not\subseteq C_a(L_a^S \setminus \{i\})\}|.$$

The full frontier vector is the preservation gate; the main figure reports its weighted scalar change. Operational contribution and frontier sparsity are post-compression, validation-derived descriptions. Module uniqueness and descendant dependence are post-compression structural descriptions of the final safe library and use no held-out outcome. Generative contribution is computed from held-out Q_a, and the "supports best descendant" flag also uses the ex post identity of the maximizing candidate. None of these policy-characteristic statistics enters pruning.

S6.4 Primary safe-versus-frontier-only audit

Frontier-only pruning reduces the libraries from 80 to 3 and from 202 to 5. It leaves every current cell frontier unchanged, yet $\Delta Q_{\mathrm{T}}(L_{\mathrm{T}}^F) = -0.0016$ and $\Delta Q_{\mathrm{A}}(L_{\mathrm{A}}^F) = -0.1020$; both changes are ex post. Innovation-safe compression retains 25 and 100 policies, respectively, and satisfies

$$\widehat{F}_a(\,\cdot\,; L_a^S) = \widehat{F}_a(\,\cdot\,; L_a), \qquad C_a(L_a^S) = C_a(L_a), \qquad Q_a(L_a^S) = Q_a(L_a).$$

The frontier-only libraries retain 12 of 38 and 14 of 113 source-closure modules, whereas the innovation-safe closure identities are $38 = 38$ and $113 = 113$. Thus the financial audits show, using held-out outcomes, that present-frontier preservation alone did not preserve access to the highest-quality enabled descendants in these two audits. This diagnostic statement neither changes either pruning rule nor turns realized quality into information known at deletion time.

Each safe library has exactly one positive ex post generative carrier: GLD has $(O, G) = (0, 0.0016)$ and UNG has $(O, G) = (0, 0.0534)$. Both uniquely support the audit's best enabled descendant. With only two carriers, the structural patterns remain descriptive.

Design or diagnostic	Locked terminal audit	Annual walk-forward audit
Universe and grammar	25 surviving ETFs; 3 belief states; 2,400 strategies	100 endpoint-stable surviving ETFs; 5 belief states; 9,600 strategies
Split	Development 2009–2014; validation 2015–2019; locked 2020–2024	Same initial split; five trailing-five-year decisions for 2020–2024
Timing and costs	Signal through close t; two-return-index lag; 5 bp one-way base cost	Same lag and 5 bp base cost; 1 and 10 bp sensitivities in both runs
Pruning information	Validation frontier; innovation-safe also uses structural closure	Validation frontier; innovation-safe also uses structural closure
Ranking information	Development/validation score fixed after compression and before locked outcomes	Trailing-through-$y-1$ score fixed after compression and before target y
Frontier-only pruning	$80 \to 3$; current change 0; ex post $\Delta Q_a = -0.0016$	$202 \to 5$; current change 0; ex post $\Delta Q_a = -0.1020$
Innovation-safe pruning	$80 \to 25$; frontier and closure changes 0; ex post $\Delta Q_a = 0$	$202 \to 100$; frontier and closure changes 0; ex post $\Delta Q_a = 0$
Q_a information status	Ex post held-out outcome; not an input, forecast, policy score, or deployable criterion	Ex post held-out outcome; not an input, forecast, policy score, or deployable criterion
Post-decision diagnostics	Compression, uniqueness, and dependence are retrospective; locked utility is held out	Compression, uniqueness, and dependence are retrospective; realized coverage and oracle regret are held out
Ex post generative carrier	GLD: operational 0, generative 0.0016	UNG: operational 0, generative 0.0534
Coverage outcome	Top-10 improvement -0.1560; best comparator -0.0685	Mean top-5 set coverage 0.0704; best comparator 0.0119; positive in 3/5 years

Table 17: Locked financial designs, compression diagnostics, and secondary coverage outcomes. Pruning uses the validation frontier and, for innovation-safe deletion, structural closure. Compression and policy characteristics are post-pruning descriptions. Coverage scores are fixed from validation or pre-target data after compression. The Q_a entries and displayed performance outcomes use held-out data and are not pruning inputs, forecasts, policy scores, or deployable criteria. The audits use different held-out estimands.

S6.5 Secondary coverage-ranking definitions

Coverage scores are post-compression ranking inputs, never pruning inputs. The terminal score uses validation gaps and a development-estimated transition; each annual score uses data only through $y-1$. Selected sets are hashed before outcomes. Locked utility and annual realized coverage are held out, rank associations and intervals are retrospective, and greedy-oracle regret uses a target-year oracle.

For the secondary terminal audit, method m selects a frozen set $S_{\mathrm{T}m}$ of $k = 10$ candidates. With locked net utility $u_{\mathrm{T}}(c)$, incumbent benchmark $u^0_{\mathrm{T}} := \max_{i \in L_{\mathrm{T}}} u_{\mathrm{T}}(i)$, and prespecified score $x_{\mathrm{T}c}(m)$, define

$$R_{\mathrm{T}}(m) := \frac{1}{k} \sum_{c \in S_{\mathrm{T}m}} [u_{\mathrm{T}}(c) - u^0_{\mathrm{T}}], \qquad \rho_{\mathrm{T}}(m) := \mathrm{Corr}\Big(\mathrm{rank}_c\, x_{\mathrm{T}c}(m), \mathrm{rank}_c[u_{\mathrm{T}}(c) - u^0_{\mathrm{T}}]\Big).$$

Here $x_{\mathrm{T}c}(m)$ is computed from development/validation information after compression and fixed in the decision hash. Locked net utility, selected-set payoff, and rank association are held-out or retrospective audit quantities; none is available to pruning.

For annual target y, realized belief weight $\widehat{\omega}_y(b)$, and candidate gap $\widehat{g}_{cy}(b) := [\widehat{j}_{cy}(b) - \widehat{F}_y(b; L_{\mathrm{A}})]_+$, define

$$Z_y(S) := \sum_b \widehat{\omega}_y(b) \max_{c \in S} \widehat{g}_{cy}(b), \qquad R_{\mathrm{A}}(m) := \frac{1}{5} \sum_{y=2020}^{2024} Z_y(S_{\mathrm{A}my}).$$

Let $z_{cy} := \sum_b \widehat{\omega}_y(b)\widehat{g}_{cy}(b)$, x_{cmy} be the outcome-blind score, and $S_y^\star$ the target-year greedy set. Then

$$\rho_{\mathrm{A}}(m) := \frac{1}{5}\sum_y \mathrm{Corr}(\mathrm{rank}_c\, x_{cmy}, \mathrm{rank}_c\, z_{cy}), \qquad G_{\mathrm{A}}(m) := \frac{1}{5}\sum_y [Z_y(S_y^\star) - Z_y(S_{\mathrm{A}my})].$$

The score x_{cmy} and selected set $S_{\mathrm{A}my}$ use information only through $y-1$ and are fixed before the year-y decision hash. The realized occupation, gaps, individual and set coverage, rank association, and resampling inputs use target-year outcomes and are available only afterward. The set $S_y^\star$ is a target-year greedy oracle, so G_{A} is an infeasible oracle-comparator diagnostic rather than a deployable score. Finally, $I_{\mathrm{A}}(m)$ is the 2.5th–97.5th percentile of 2,000 seeded resamples of the five annual values $Z_y(S_{\mathrm{A}my})$.

S6.6 Secondary results

In the terminal audit, the coverage-ranked set has $R_{\mathrm{T}}(\text{coverage}) = -0.1560$, compared with -0.0685 for the best prespecified comparator. Candidate-level Spearman association is -0.0982, and the coverage ordering remains adverse under the locked 1, 5, and 10 basis-point schedules.

Annual marginal coverage is (0.0704), versus (0.0119) for the best comparator, with three of five positive values. Mean Spearman association is 0.1279, mean greedy-oracle regret is (0.315), and the annual-unit interval is $[0.0190, 0.1187]$. The terminal result is adverse and the annual result supportive but noisy; neither identifies expected return or market alpha.

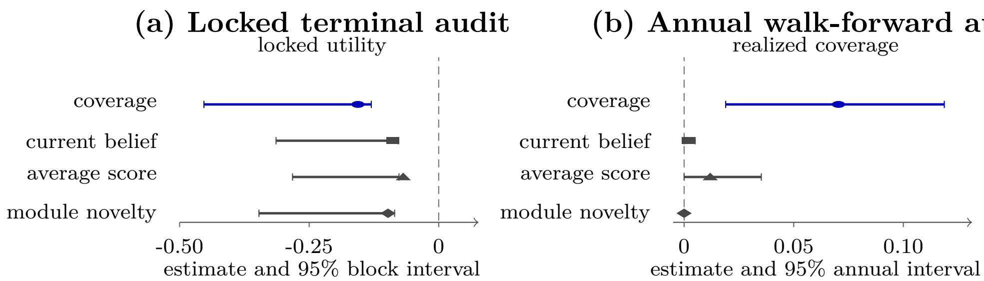

Figure 3: Secondary coverage-ranking estimands on separate audit-specific axes: held-out locked-utility change in the terminal audit and next-year realized set coverage in the annual audit. Points and 95% intervals are rendered floating-point summaries from the committed resampling rows, not exact rationals. The terminal outcome is adverse; the annual outcome is supportive but noisy, with only five annual target units. The descriptive intervals do not cover design uncertainty.

These audits are retrospective mechanism evidence. They support no causal, forecasting, alpha, or deployable-performance claim.

S7 Reproducibility Records

This section records the execution boundaries, configuration histories, hashes, and complete artifact locations for Supplements S4–S6. The commands are reproduction instructions only; compiling this supplement does not run Julia or regenerate any committed result. The public repository is available at

`https://github.com/dramirez37/financial-strategy-library-compression`.

S7.1 Canonical and presentation records

The canonical solve is deterministic and uses no random seed:

```
julia --project=julia \
  julia/scripts/solve_unified_canonical_benchmark.jl
```

The manuscript presentation layer is Julia-generated:

```
julia --project=julia \
  julia/scripts/generate_manuscript_numerical_artifacts.jl
julia --project=julia \
  julia/scripts/generate_manuscript_numerical_artifacts.jl --check
```

The first command validates the committed convergence, compression, decomposition, geometry, grammar, mechanism, and uncertainty tables before writing manuscript artifacts. The second is nonmutating and rejects byte drift. Complete canonical numerical outputs are listed in Supplement S4.

S7.2 Randomized-study configuration history

The registered v2 experiment uses the fixed maximum $N = 1024$, master seed `6075990691714899803`, and the complete trial and derived-seed registry at

`experiments/randomized_library_v2/TRIAL_REGISTRY.csv`.

The initial design, sequential-precision amendment, and executable-protocol amendment are retained at

`experiments/randomized_library_v2/DESIGN_LOCK.json`,
`experiments/randomized_library_v2/DESIGN_LOCK_AMENDMENT_1.json`,
`experiments/randomized_library_v2/DESIGN_LOCK_AMENDMENT_2.json`.

Their aggregate hashes are, respectively,

```
0b012dfc14f4ea57b0d34877a68d9a546cd499d2270903e772d78b21425d14db
d9122d258b8fe62d5872c41b0a418b1351a5798b25ddb99ec73b726e324d244f
8c278c07d998ba118d98c78cc1373a47ab63127f00d606c0042c006dac11e7be
```

The registered run, byte-drift replay, and independent result-reader audit are:

```
julia --project=julia \
  julia/scripts/run_randomized_library_stress_v2.jl
julia --project=julia \
  julia/scripts/run_randomized_library_stress_v2.jl --check
julia --project=julia \
  julia/scripts/audit_randomized_library_v2_results.jl
```

All theorem-facing calculations and point estimates use `Rational{BigInt}`. Decimal Wilson intervals, Monte Carlo standard errors, and SVG coordinates are presentation diagnostics only. The complete cumulative source is

`experiments/results/summaries/randomized_library_v2_stability_summary.csv`, and the 896-row two-level slice table is

`experiments/results/summaries/randomized_library_v2_stability_factor_summary.csv`.

The data-linked stabilization plot is

`manuscript/figures/randomized_library_v2_stability.svg`.

The exact final counts are 398 positive frontier-only losses, zero positive innovation-safe losses, 388 silent generative assets among 5,120 source assets, 256 substitutions, 141 complementarities, and 627 zero interactions. All interaction counts use four raw witnesses. These diagnostics describe Monte Carlo precision within the registered finite generator, not uncertainty about a real population.

The separately locked resource extension reuses only the frozen v2 registry. Its exact run and byte-drift check are

```
julia --project=julia \
  julia/scripts/run_randomized_library_optimization_extension.jl
julia --project=julia \
  julia/scripts/run_randomized_library_optimization_extension.jl --check
```

Every trial enumerates all 32 inactive-containing sublibraries. The readable exact summary is

`RANDOMIZED_LIBRARY_OPTIMIZATION_EXTENSION_V1.md`.

The parent report and frozen pilot remain separately indexed by `RANDOMIZED_LIBRARY_REPORT_V2.md` and `RANDOMIZED_LIBRARY_REPORT.md`; they are not pooled.

S7.3 Financial configuration and amendment history

Both financial audits use the licensed CRSP/WRDS snapshot associated with source-repository commit `3146029aa71dfe7639ede9ca79f81602165125e8`. Preparation scripts perform no download. The terminal audit uses seed `6075990691714899801`; the annual walk-forward audit uses seed `6075990691714899802`. Their ignored panels contain 106,975 and 427,900 rows, each with 4,279 complete common dates. Raw and row-level data are nonredistributable.

The terminal configuration, audit, results, and limitations records are

`experiments/configs/financial_terminal_audit.toml`,
`experiments/financial_terminal_audit/DATA_AUDIT.md`,
`experiments/financial_terminal_audit/DRAFT_RESULTS.md`,
`experiments/financial_terminal_audit/EMPIRICAL_LIMITATIONS.md`.

The annual walk-forward audit's initial lock, six retained amendments, amendment history, final lock, frozen universe, audit, results, and limitations are

`experiments/financial_annual_walkforward_audit/DESIGN_LOCK_INITIAL.json`,
`experiments/financial_annual_walkforward_audit/DESIGN_LOCK_AMENDED_1.json` through
`experiments/financial_annual_walkforward_audit/DESIGN_LOCK_AMENDED_6.json`,
`experiments/financial_annual_walkforward_audit/DESIGN_AMENDMENTS.md`,
`experiments/financial_annual_walkforward_audit/DESIGN_LOCK.json`,
`experiments/financial_annual_walkforward_audit/universe_selection.toml`,
`experiments/financial_annual_walkforward_audit/DATA_AUDIT.md`,
`experiments/financial_annual_walkforward_audit/DRAFT_RESULTS.md`,
`experiments/financial_annual_walkforward_audit/EMPIRICAL_LIMITATIONS.md`.

The initial protocol, sparse-support amendment, and report-only corrections remain in that versioned history. Reruns must preserve negative and mixed outcomes unless a new experiment is registered.

The cross-audit financial resource optimization is a separate exact post-solve analysis of the two committed aggregate audit records; it is not a third licensed-data audit and does not read row-level inputs. Its nonmutating lock and certificate checks are

```
julia --project=julia \
  julia/scripts/lock_financial_resource_optimization.jl --check
julia --project=julia \
  julia/scripts/verify_financial_resource_optimization_outputs.jl
```

The registered design and readable exact report are

`experiments/financial_resource_optimization/DESIGN_LOCK.json`,
`FINANCIAL_RESOURCE_OPTIMIZATION.md`.

S7.4 Complete artifact index

`ARTIFACT_MANIFEST.md` is the authoritative configuration, data, decision, RNG, output-hash, and correction-history index. `REPRODUCIBILITY.md` records the complete command surface and component-specific gates. `EMPIRICAL_INFORMATION_SET_AUDIT.md` records the code-level financial execution order and quantity-by-quantity information classification.

Within `experiments/results/summaries/`, the following prefix families exhaust the study records used in the main paper and Supplements S5–S6:

Registered parent study. The parent-study prefix is

`randomized_library_v2_*`.

It excludes the separately registered optimization-extension prefix below. This family contains trials, assets, profiles, modules, closures, kernels, projects, pruning, actions, raw-witness manifests, rectangle corners and transitions, interaction signs, method, factor, relationship, stability, and summary records.

Randomized resource extension. `randomized_library_v2_optimization_v1_*`. This family contains trial, sublibrary, compression, capacity, breakpoint, penalized, elasticity, factor-summary, and aggregate-summary records.

Exact response experiments. `unified_comparative_statics_*` and `theorem_mechanism_*`. These families contain exact fixtures, the 2,430-row interaction response, sign checks, policy-map rows, decomposition, pruning-loss, coverage, selection, metadata, and aggregate summaries.

Locked terminal financial audit. `financial_terminal_audit_*`. This namespace is disjoint from the annual walk-forward prefix below. The family contains grammar, candidate, pruning, ranking, decomposition, uncertainty, cost-sensitivity, mechanism, figure-source, and status records.

Annual walk-forward financial audit. `financial_annual_walkforward_audit_*`. This family contains grammar, candidate, pruning, selection, episode-ranking, ranking, decomposition, uncertainty, cost-sensitivity, mechanism, universe-figure, and status records.

Cross-audit financial resource optimization. `financial_resource_optimization_*`. This family contains registered weights, selected-library memberships, exact frontier certificates, solution summaries, and status metadata derived from the two aggregate audit records.

The corresponding supplement figures are

`manuscript/figures/dynamic_research_policy_regions.tex`,
`manuscript/figures/randomized_library_v2_stability.svg`,
`manuscript/figures/financial_coverage_comparison.tex`.

The theorem ledger, formal correspondence, and validation matrix retain mathematical validity, Lean kernel verification, Julia validation, and empirical relevance as separate evidence dimensions.